

Finite-time perturbation selection in evolving HCP crystal plasticity: path-dependent direction, wavelength and mechanism

[Author names and affiliations to be supplied]

Abstract

Frozen adiabatic-shear analyses rank instantaneous roots but cannot identify which crystal-resolved disturbance wins along an evolving loading path. We pose this finite-time selection problem for an 18-slip HCP crystal. Fourier linearization gives an 84-coordinate differential–algebraic descriptor; exact elimination of 18 micromorphic fields yields a nonsingular 69-state generator. A time-ordered propagator selects direction and wavenumber for seven input–output maps. Stationary and commuting-normal controls recover frozen-spectrum equality, whereas HCP paths depart through non-normality, eigenvector rotation and chronology. Re-optimization under five norms separates compact-domain elastic waves from interior constitutive-localization maxima; the terminal constitutive map selects $k = 6.84 \times 10^3 \text{ m}^{-1}$ and converged gain 6.29×10^5 . Same-selector piecewise-frozen products converge to the ordered map. Three added rate/temperature paths and four local parameter perturbations preserve the terminal basin but expose scale sensitivity. A four-mode Galerkin replay reproduces both optimized vectors at small amplitude, but full-Fourier nonlinear continuation becomes non-finite. Synchronized Grade-II titanium data retain only qualitative stress-before-temperature ordering, not timing or magnitude. The framework therefore closes finite-time linear selection; mature width, calibration and propagation resistance remain open.

Keywords: adiabatic shear localization, HCP crystal plasticity, finite-time propagator, non-normal amplification, dislocation density, wavelength selection

1. Introduction

Adiabatic shear localization is usually introduced as a competition between work hardening, rate sensitivity and thermal softening. Classical perturbation theories sharpened this picture by adding inertia and thermal diffusion and by asking when a homogeneous simple-shear solution admits a growing disturbance [1, 2, 3, 4, 5]. Subsequent analyses clarified thermomechanical coupling, loss of stability and the role of regularizing physics in pattern selection [6, 7, 8]. In their natural domain, these theories answer an indispensable question: which root of a scalar, frozen dispersion relation has the largest instantaneous real part? They do not, however, determine which perturbation of an anisotropic crystal accumulates the largest observable amplification over an interval during which the tangent, active slip systems, dislocation populations and temperature all evolve.

That distinction is physical rather than semantic. HCP crystals possess competing basal, prismatic and pyramidal slip systems whose resolved driving forces depend on orientation. Crystal-plasticity calculations show that slip competition, texture and microstructural heterogeneity strongly affect localization [9, 10, 11]. Thermomechanically coupled formulations add heat generation and evolving resistance [12, 13]; micromorphic and strain-gradient formulations add internal fields, grain-boundary interactions and length-dependent recovery [14, 15, 16, 17, 18, 19]. The resulting perturbation operator is both high dimensional and nonautonomous. Its eigenvectors can rotate, its branches can exchange dominance, and non-normal interactions can amplify a disturbance that is not represented by the leading frozen eigenvector at any single time.

The unresolved question is therefore specific: *for an orientation-dependent, multi-slip, dislocation-thermal HCP crystal whose base state continues to evolve, which initial disturbance obtains the largest cumulative gain over the complete loading history, and why do its direction, wavenumber and state composition change with the observation horizon?* Integrating the largest frozen growth rate is not an answer unless the evolving operators and their modal subspaces satisfy restrictive commutation and compatibility conditions. Those conditions are neither guaranteed by a positive frozen eigenvalue nor tested by a conventional dispersion plot.

There is also a structural obstacle. With displacement inertia, temperature and internal variables as differential coordinates but micromorphic slip fields as algebraic coordinates, the Fourier-linearized crystal problem is a

quadratic differential–algebraic descriptor. Adding fictitious storage to the algebraic fields creates artificial fast modes; eliminating them without a determinant and reconstruction audit can silently delete or move physical roots. A defensible finite-time theory must first produce an ordinary generator that is exactly equivalent on the finite spectrum and only then define an observable propagator.

This paper resolves those two levels in sequence (Fig. 1). First, an 84-coordinate HCP descriptor is reduced exactly to a 66-coordinate quadratic pencil and a 69-state generator, with the eliminated 18-coordinate microfield reconstructed algebraically. Second, the chronological propagator is evaluated in a fixed physical scaling. A projected constitutive input–output gain defines the set-valued direction–wavenumber localization selector. The central frozen-spectrum test is deliberately separate: it compares the complete 69-state propagator norm with the full-state spectral abscissa on the same branch, base history and norm. This avoids assigning a full-state frozen rate to a projected subspace that is not dynamically invariant.

Six additional evidence layers are used. The generator is checked against an independent generalized Schur decomposition of the augmented descriptor. Stationary-normal, commuting-normal and invariant-projection controls test when frozen accumulation is legitimate. The discrimination is repeated for three independently selected crystal orientations, while commutators, modal condition numbers, eigenvector rotation and order-reversed propagation diagnose its origin. Eleven norm definitions and seven selector pairs test conditioning of the localization result. The direction–wavenumber basin is challenged by nested Sobol reconnaissance, explicit prior-branch anchors, independent refinements and 129-state verification; because the blind design misses the terminal basin, this is retained only as an anchor-assisted basin audit. Finally, the actual near-onset and terminal singular vectors are advanced in a declared four-mode nonlinear Galerkin replay, while a separate all-mode audit and broader seed-spectrum/grid gates test—and presently reject—complete nonlinear and material-width closure.

The scope is deliberately bounded. The scalar Bai implementation is an independently tested classical anchor, but a strict analytical HCP-to-Bai limit is not claimed. The search audit demonstrates retention of known basins, not independent global discovery. The critical time is conditioned on a fixed norm, selectors and threshold, and the pure-titanium parameters are literature constrained rather than specimen identified. The nonthermal power remainder is a passive ledger rather than an identified dislocation free en-

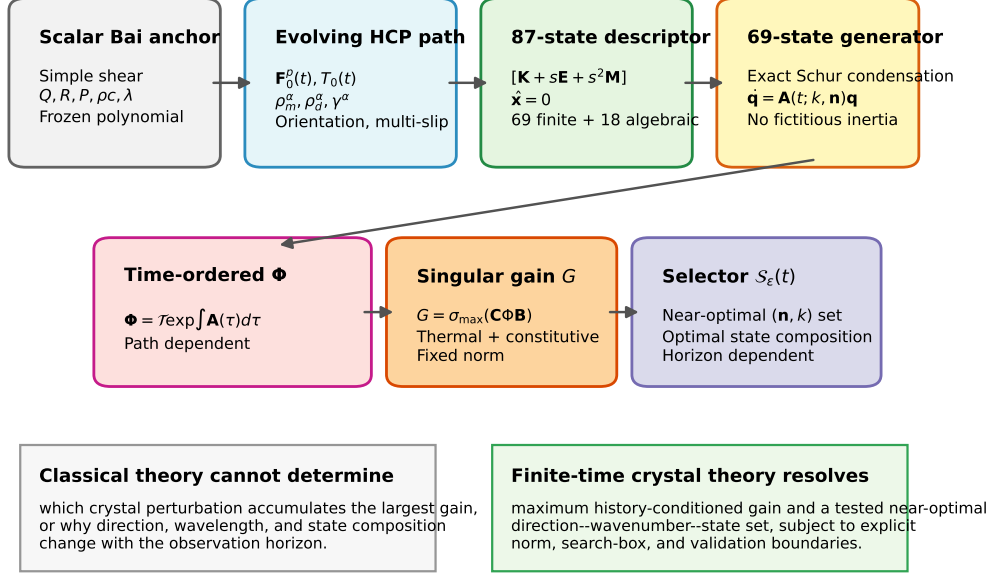


Figure 1: Theory and variable framework. The classical scalar Bai problem provides the instantaneous thermoplastic anchor; the HCP formulation resolves orientation, 18 slip systems, dislocation populations, temperature, inertia and algebraic micromorphic fields; exact descriptor reduction then permits time-ordered finite-time selection. The strict analytical HCP-to-Bai limit remains an explicit open gate.

ergy. These boundaries allow the paper to make one strong claim: a crystal-resolved, nonautonomous propagator answers a perturbation-selection question that frozen theories cannot answer, while keeping nonlinear length-scale, microstructural-energy and propagation-resistance closure as open problems.

Nomenclature

Symbol	Definition
$\mathbf{F}, \mathbf{F}_e, \mathbf{F}_p$	total, elastic, and plastic deformation gradients
$\mathbf{P}$	first Piola–Kirchhoff stress
$\gamma_\alpha, \dot{\gamma}_\alpha$	signed slip and slip rate on system α
$\mathbf{s}_\alpha, \mathbf{m}_\alpha$	slip direction and reference slip-plane normal

Symbol	Definition
T, ρ_0, c, κ	temperature, reference density, specific heat, and conductivity tensor
$\zeta \in \mathbb{R}^{18}$	micromorphic slip field
$\mathbf{a} \in \mathbb{R}^{62}$	independent local constitutive coordinates
$\mathbf{z}_{84}$	augmented Fourier amplitude $[\mathbf{u}_3, T, \zeta_{18}, \mathbf{a}_{62}]$
$\mathbf{y}_{66}$	retained quadratic-pencil coordinates $[\mathbf{u}_3, T, \mathbf{a}_{62}]$
$\mathbf{d}_{63}$	differential coordinates $[T, \mathbf{a}_{62}]$
$\mathbf{q}_{69}$	generator state $[\mathbf{u}_3, \mathbf{v}_3, \mathbf{d}_{63}]$
$\mathbf{K}, \mathbf{E}, \mathbf{M}$	stiffness, first-order storage, and inertia blocks
$\mathcal{Q}(s)$	84-coordinate quadratic matrix pencil
$\mathbf{K}_{\zeta\zeta}$	algebraic microfield block
$\mathbf{K}_{\zeta-\text{red}}^{(66)}$	exact Schur-reduced retained stiffness
$\mathbf{R}, \mathbf{C}$	mechanical inertia and differential storage blocks
$\mathbf{A}_{\text{dyn}}$	nonsingular 69-state time-dependent generator
$\Phi(t, t_0)$	finite-time state-transition operator
$\mathbf{D}_q$	fixed diagonal physical scaling of generator coordinates
$\mathbf{P}_{\text{in}}, \mathbf{P}_{\text{out}}$	prescribed input and output selectors
$G, G_\star$	selected input–output gain and its direction–wavenumber envelope
G_D^{full}	full-state gain $\ \mathbf{D}_q^{-1}\Phi\mathbf{D}_q\ _2$ with identity selectors
α, μ_2	spectral abscissa and Euclidean logarithmic norm of the same scaled full-state generator
$\Delta_{\text{FT}}^{\text{full}}$	$\log G_D^{\text{full}} - \int \alpha dt$ on the same state space and norm
$\kappa_\lambda, \theta_\lambda$	instantaneous leading-mode condition number and adjacent leading-eigenvector rotation
t_c	first threshold-crossing time for the locked norm and selectors
$\mathbf{n}, k$	propagation direction and wavenumber
Ω	admitted projective direction–wavenumber search domain
η_b^{out}	scaled finite-time output fraction assigned to mechanism block b
λ_{FT}	selected finite-time wavelength $2\pi/k^*$
$\mathcal{P}_{\text{cp}}, r_T, u_s$	crystal-plastic power, heat-source rate and passive nonthermal ledger coordinate

Symbol	Definition
ψ_ζ	micromorphic penalty and gradient contribution to the free energy
w_{FWHM}^X	nonlinear FWHM of observable X
C_{FWHM}	ratio $w_{\text{FWHM}}^X/\lambda_{\text{FT}}$ for stated windows

2. HCP base model and dynamic descriptor

2.1. Finite-strain crystal state and explicit local closure

The homogeneous crystal follows the multiplicative split

$$\mathbf{F} = \mathbf{F}_e \mathbf{F}_p, \quad \det \mathbf{F}_p = 1, \quad \mathbf{L}_p = \dot{\mathbf{F}}_p \mathbf{F}_p^{-1} = \sum_{\alpha=1}^{18} \dot{\gamma}^\alpha \mathbf{s}^\alpha \otimes \mathbf{m}^\alpha. \quad (1)$$

The HCP geometry contains three basal $\langle a \rangle$, three prismatic $\langle a \rangle$ and twelve pyramidal $\langle c + a \rangle$ systems. Twinning is inactive in the present baseline. The 62 independent local coordinates are

$$\mathbf{a} = [\boldsymbol{\xi}_8, \boldsymbol{\rho}_m^{18}, \boldsymbol{\rho}_d^{18}, \boldsymbol{\gamma}^{18}]^\top, \quad (2)$$

where $\boldsymbol{\xi}$ is an eight-coordinate chart of the isochoric plastic distortion, ρ_m^α and ρ_d^α are mobile and dipole dislocation densities, and γ^α are signed slips. Imposing the chart before linearization removes the redundant plastic-volume coordinate.

For slip system α , forest density, slip resistance and mean free path are

$$\rho_f^\alpha = \max \left[\sum_{\beta=1}^{18} h_{\alpha\beta} (\rho_m^\beta + \rho_d^\beta), \rho_{\min} \right], \quad (3)$$

$$g^\alpha = \tau_0^\alpha + a_T \mu(T) b^\alpha \sqrt{\rho_f^\alpha}, \quad (4)$$

$$\Lambda^\alpha = \left[d_g^{-1} + \frac{\sqrt{\rho_f^\alpha}}{K_\Lambda} \right]^{-1}. \quad (5)$$

The resolved Mandel stress τ^α is coupled to the micromorphic slip ζ^α through

$$\tau_{\text{eff}}^\alpha = \tau^\alpha + H_\chi (\zeta^\alpha - \gamma^\alpha). \quad (6)$$

On an active smooth branch, the thermally activated kinetic law implemented in the base model is

$$\dot{\gamma}^\alpha = \rho_m^\alpha b^\alpha v_0^\alpha \exp \left[-\frac{\Delta F^\alpha}{k_B T} \langle 1 - \bar{\tau}_\alpha^{p^\alpha} \rangle^{q^\alpha} \right] \text{sign}(\tau_{\text{eff}}^\alpha), \quad (7)$$

$$\bar{\tau}_\alpha = \min \left(\frac{\langle |\tau_{\text{eff}}^\alpha| - g^\alpha \rangle}{\tau_{\text{cut}}^\alpha}, 1 \right), \quad \dot{\gamma}^\alpha = 0 \quad \text{if } |\tau_{\text{eff}}^\alpha| \leq g^\alpha. \quad (8)$$

Here $\langle x \rangle = \max(x, 0)$. The active-set condition in Eq. (8), the density floor and the activation cap are recorded before differentiation; a perturbation probe that changes one of these branches is rejected.

The continuous-rate counterparts of the implemented semi-implicit density update can be written

$$\dot{\rho}_m^\alpha = S^\alpha - (F^\alpha + G^\alpha) \rho_m^\alpha, \quad S^\alpha = \frac{|\dot{\gamma}^\alpha|}{b^\alpha \Lambda^\alpha}, \quad (9)$$

$$\dot{\rho}_d^\alpha = F^\alpha \rho_m^\alpha - (G^\alpha + C^\alpha) \rho_d^\alpha, \quad (10)$$

where

$$d_{\text{check}}^\alpha = c_d b^\alpha, \quad d_{\text{hat}}^\alpha = \min \left[\max \left(\frac{3\mu_0 b^\alpha}{16\pi |\tau_{\text{eff}}^\alpha|}, d_{\text{check}}^\alpha \right), \Lambda^\alpha \right], \quad (11)$$

$$F^\alpha = \frac{2(d_{\text{hat}}^\alpha - d_{\text{check}}^\alpha)_+}{b^\alpha} |\dot{\gamma}^\alpha|, \quad G^\alpha = \frac{2d_{\text{check}}^\alpha}{b^\alpha} |\dot{\gamma}^\alpha|, \quad (12)$$

$$C^\alpha = \nu_c \exp[-Q_c/(k_B T)]. \quad (13)$$

The effective crystal-plastic power and its registered thermal/nonthermal partition are

$$\mathcal{P}_{\text{cp}} = \sum_{\alpha=1}^{18} \tau_{\text{eff}}^\alpha \dot{\gamma}^\alpha \geq 0, \quad r_T = \beta \mathcal{P}_{\text{cp}}, \quad \dot{u}_s = (1 - \beta) \mathcal{P}_{\text{cp}}, \quad 0 \leq \beta \leq 1. \quad (14)$$

Here u_s is a passive nonthermal configurational ledger. It is not identified with a dislocation free energy, generates no hardening force and is not an active coordinate of the perturbation generator. The numerical base path audits the Mandel/slip power equality and the mechanical–microforce driving-power ledger.

Equations (1)–(14), anisotropic thermoelasticity and the local micromorphic force define

$$(\mathbf{P}, r_T, \mathbf{q}, \dot{\mathbf{a}}) = \mathcal{R}(\mathbf{F}, T, \boldsymbol{\zeta}, \mathbf{a}). \quad (15)$$

The perturbation operator uses its continuous, branch-locked tangent

$$\mathrm{d}\mathcal{R} = \mathcal{R}_F : \mathrm{d}\mathbf{F} + \mathcal{R}_T \mathrm{d}T + \mathcal{R}_\zeta \mathrm{d}\boldsymbol{\zeta} + \mathcal{R}_a \mathrm{d}\mathbf{a}, \quad (16)$$

not an endpoint algorithmic tangent. Equation (16) is the precise interface by which anisotropic elasticity, slip, dislocation evolution, heating and micromorphic coupling enter the descriptor.

2.2. Recoverable micromorphic energy, entropy production and loading path

The recoverable micromorphic contribution is

$$\psi_\chi = \frac{1}{2} H_\chi \|\boldsymbol{\zeta} - \boldsymbol{\gamma}\|^2 + \frac{1}{2} \text{Grad } \boldsymbol{\zeta} : \mathbf{H}_\nabla : \text{Grad } \boldsymbol{\zeta}, \quad \mathbf{H}_\nabla = \mathbf{H}_{\text{slip}} + \mathbf{H}_{\text{Nye}}, \quad (17)$$

$$\mathbf{H}_{\text{Nye}} = \mu \ell_N^2 \mathbf{C}_N^H \mathbf{C}_N. \quad (18)$$

Thus the energetic microstress and higher-order stress are

$$\boldsymbol{\pi} = H_\chi (\boldsymbol{\zeta} - \boldsymbol{\gamma}), \quad \boldsymbol{\xi} = \frac{\partial \psi_\chi}{\partial \text{Grad } \boldsymbol{\zeta}}, \quad \frac{\partial \psi_\chi}{\partial \boldsymbol{\gamma}} = -\boldsymbol{\pi}. \quad (19)$$

The last identity explains the effective driving stress $\boldsymbol{\tau}_{\text{eff}} = \boldsymbol{\tau} + \boldsymbol{\pi}$: after the recoverable micromorphic-energy rate is removed from mechanical slip power, the distributable remainder is $\mathcal{P}_{\text{cp}}$ in Eq. (14). Under periodic or homogeneous microtraction boundary conditions, the local split gives

$$\mathcal{D}_{\text{mat}} = \mathcal{P}_{\text{cp}} - \dot{u}_s = r_T \geq 0, \quad \dot{\mathcal{S}}_{\text{prod}} = \frac{r_T}{T} + \frac{\text{Grad } T \cdot \boldsymbol{\kappa} \text{Grad } T}{T^2} \geq 0. \quad (20)$$

The associated global first-law ledger contains kinetic energy, thermoelastic internal energy, ψ_χ and u_s , with external mechanical power, microtraction power and boundary heat flux. This closes the declared power/heat book-keeping, but it deliberately does not claim that u_s is a recoverable dislocation energy or that the density variables possess an identified Helmholtz potential. A constitutively coupled dislocation-storage law remains outside the present model.

The slip-gradient part of $\mathbf{H}_\nabla$ is full rank for the adopted 18-slip geometry; the Nye part is rank deficient and cannot alone control every micromorphic direction. In the reference configuration,

$$\rho_0 \ddot{\mathbf{u}} - \text{Div } \mathbf{P} = \mathbf{0}, \quad (21)$$

$$\rho_0 c \dot{T} - \text{Div}(\boldsymbol{\kappa} \text{Grad } T) - r_T = 0, \quad (22)$$

$$\mathbf{q} - \text{Div}(\mathbf{H}_\nabla \text{Grad } \boldsymbol{\zeta}) = \mathbf{0}, \quad (23)$$

$$\dot{\mathbf{a}} - \mathbf{f}_a(\mathbf{F}, T, \boldsymbol{\zeta}, \mathbf{a}) = \mathbf{0}. \quad (24)$$

The homogeneous base path is simple shear,

$$\mathbf{F}_0(t) = \mathbf{I} + \gamma_0(t) \mathbf{e}_1 \otimes \mathbf{e}_2, \quad \dot{\gamma}_0 = 10^4 \text{ s}^{-1}, \quad T_0(0) = 293.15 \text{ K}, \quad (25)$$

integrated to $\gamma_0 = 1.4$ at $140 \mu\text{s}$. It is a reproducible single-crystal research baseline, not a specimen-specific calibration.

2.3. Fourier-linearized descriptor

At a frozen base time, use

$$\delta \mathbf{y}(\mathbf{X}, t) = \hat{\mathbf{y}} \exp(st + ik\mathbf{n} \cdot \mathbf{X}), \quad k > 0, \quad \mathbf{n} \in \mathbb{S}^2. \quad (26)$$

In the ordering $[\hat{\mathbf{u}}_3, \hat{T}, \hat{\boldsymbol{\zeta}}_{18}, \hat{\mathbf{a}}_{62}]$, linearization gives

$$\underbrace{[\mathbf{K}(t; k, \mathbf{n}) + s\mathbf{E}(t) + s^2\mathbf{M}(t)]}_{\mathcal{Q}(s; t, k, \mathbf{n})} \hat{\mathbf{z}}_{84} = \mathbf{0}. \quad (27)$$

Only the three mechanical coordinates carry second-order inertia; temperature and the 62 local coordinates carry first-order storage; the 18 micromorphic coordinates are algebraic. The ik and k^2 blocks retain acoustic, thermo-mechanical, conduction and gradient couplings. Equation (27) is therefore a quadratic descriptor, not a conventional eigenproblem.

3. Exact reduction and finite-time perturbation theory

3.1. Exact dynamic reduction and finite-time perturbation selection

This subsection closes the theoretical gap between the frozen-time quadratic descriptor in Eq. (27) and an observable perturbation amplified over a finite, evolving loading history. The construction is for the single-crystal HCP base state. Experimental parameter identification and independent full-field material validation are outside the present scope.

3.1.1. Admissible base history and operator hypotheses

Let $I = [t_0, t_f]$, let $k > 0$, and let $\mathbf{n} \in \mathbb{S}^2$. The following hypotheses define the domain on which the reduction is asserted.

- (H1) The homogeneous base history is continuous and piecewise C^1 , remains on one recorded smooth constitutive branch on I , and supplies continuous tangent blocks (piecewise continuity suffices for propagation, but not for every threshold-continuity statement below).
- (H2) The density and heat capacity are strictly positive, the continuous tangent blocks are finite and bounded, and the mechanical inertia and differential storage blocks retain constant rank on I .
- (H3) The 18-by-18 algebraic microfield block is uniformly invertible: $\inf_{t \in I} \sigma_{\min}(\mathbf{K}_{\zeta\zeta}(t; k, \mathbf{n})) > 0$ in the locked assembled physical coordinates for every admitted $(k, \mathbf{n})$; the infimum is also uniform over each compact parameter domain used below.
- (H4) The eight-coordinate $\text{SL}(3)$ chart is imposed before a spectrum or propagator is interpreted. The continuous local symbol does not impose finite-cell wavevector quantization; rigid translation is projected only in the $k = 0$ limit. A periodic finite cell instead supplies a separate discrete Bloch/Floquet wavevector set and its own gauge treatment.
- (H5) State scales, input and output selectors, the gain threshold, the direction–wavenumber search domain, and all numerical tolerances are fixed before the resulting optimum is inspected.

If H1 fails at a switching surface, an ordinary Jacobian does not define the claimed operator. If H3 fails, the Schur reduction below is disabled and the augmented descriptor is retained as a possible algebraic or microforce bifurcation. Thus invertibility is an admission condition, not a conclusion manufactured by regularization.

3.1.2. The 84-coordinate quadratic pencil

For the locked ordering

$$\hat{\mathbf{z}}_{84} = \left[\hat{\mathbf{u}}_3, \hat{T}, \hat{\boldsymbol{\zeta}}_{18}, \hat{\mathbf{a}}_{62} \right]^\top, \quad (28)$$

write the dynamic Fourier operator as the quadratic matrix polynomial

$$\mathcal{Q}(s; t, k, \mathbf{n}) = \mathbf{K}(t; k, \mathbf{n}) + s\mathbf{E}(t) + s^2\mathbf{M}(t), \quad \mathcal{Q}\hat{\mathbf{z}}_{84} = \mathbf{0}. \quad (29)$$

The only nonzero block of $\mathbf{M}$ is the 3-by-3 mechanical block $\mathbf{R}$; the nonzero differential block of $\mathbf{E}$ is

$$\mathbf{C} = \text{diag}(\rho_0 c, \mathbf{I}_{62}). \quad (30)$$

No artificial inertia or first-order storage is assigned to ζ . All ik and k^2 terms, including the directional acoustic, conduction, and gradient-recovery terms, remain in $\mathbf{K}(t; k, \mathbf{n})$.

Partition

$$\hat{\mathbf{y}} = [\hat{\mathbf{u}}_3, \hat{\mathbf{d}}_{63}]^\top, \quad \hat{\mathbf{d}} = [\hat{T}, \hat{\mathbf{a}}_{62}]^\top, \quad (31)$$

and place $\hat{\zeta}$ second in the block partition. Because its rows and columns carry no time derivative,

$$\mathcal{Q} = \begin{bmatrix} \mathbf{K}_{yy} + s\mathbf{E}_{yy} + s^2\mathbf{M}_{yy} & \mathbf{K}_{y\zeta} \\ \mathbf{K}_{\zeta y} & \mathbf{K}_{\zeta\zeta} \end{bmatrix}. \quad (32)$$

Proposition 1 (exact algebraic reduction). Under H3, the algebraic amplitude and the retained stiffness are

$$\hat{\zeta} = -\mathbf{K}_{\zeta\zeta}^{-1} \mathbf{K}_{\zeta y} \hat{\mathbf{y}}, \quad (33)$$

$$\mathbf{K}_{\zeta\text{-red}}^{(66)} = \mathbf{K}_{yy} - \mathbf{K}_{y\zeta} \mathbf{K}_{\zeta\zeta}^{-1} \mathbf{K}_{\zeta y}. \quad (34)$$

Moreover,

$$\det \mathcal{Q} = \det \mathbf{K}_{\zeta\zeta} \det \left(\mathbf{K}_{\zeta\text{-red}}^{(66)} + s\mathbf{E}_{yy} + s^2\mathbf{M}_{yy} \right), \quad (35)$$

so the augmented and reduced problems have the same finite eigenvalues, including algebraic multiplicity, and every retained eigenvector reconstructs a unique 84-coordinate amplitude through Eq. (33).

Proof. The lower block row of Eq. (32) gives Eq. (33). Substitution gives Eq. (34); block Gaussian elimination gives Eq. (35). The eliminated block is independent of s , so the operation neither creates nor deletes a finite root. The complete block calculation and the converse reconstruction are recorded in Appendix Appendix A. $\square$

3.1.3. Nonsingular 69-state generator

Write $\mathbf{K}_{\zeta\text{-red}}^{(66)}$ in the $[\hat{\mathbf{u}}, \hat{\mathbf{d}}]$ ordering as $\mathbf{K}_{uu}$, $\mathbf{K}_{ud}$, $\mathbf{K}_{du}$, and $\mathbf{K}_{dd}$. The reduced time-domain equations are

$$\mathbf{R} \ddot{\hat{\mathbf{u}}} + \mathbf{K}_{uu} \hat{\mathbf{u}} + \mathbf{K}_{ud} \hat{\mathbf{d}} = \mathbf{0}, \quad (36)$$

$$\mathbf{C} \dot{\hat{\mathbf{d}}} + \mathbf{K}_{du} \hat{\mathbf{u}} + \mathbf{K}_{dd} \hat{\mathbf{d}} = \mathbf{0}. \quad (37)$$

Introduce physical velocity $\hat{\mathbf{v}} = \dot{\hat{\mathbf{u}}}$ and the state

$$\hat{\mathbf{q}} = [\hat{\mathbf{u}}_3, \hat{\mathbf{v}}_3, \hat{\mathbf{d}}_{63}]^\top \in \mathbb{C}^{69}. \quad (38)$$

Then

$$\dot{\hat{\mathbf{q}}} = \mathbf{A}_{\text{dyn}}(t; k, \mathbf{n})\hat{\mathbf{q}}, \quad \mathbf{A}_{\text{dyn}} = \begin{bmatrix} \mathbf{0} & \mathbf{I} & \mathbf{0} \\ -\mathbf{R}^{-1}\mathbf{K}_{uu} & \mathbf{0} & -\mathbf{R}^{-1}\mathbf{K}_{ud} \\ -\mathbf{C}^{-1}\mathbf{K}_{du} & \mathbf{0} & -\mathbf{C}^{-1}\mathbf{K}_{dd} \end{bmatrix}. \quad (39)$$

Proposition 2 (finite-spectrum equivalence). Under H2–H3, s is a finite root of Eq. (29) if and only if it is an eigenvalue of $\mathbf{A}_{\text{dyn}}$. Hence the admitted dynamic pencil has 69 finite roots, counted with algebraic multiplicity: six mechanical first-order coordinates and 63 thermal/internal-state coordinates.

More precisely, with $\mathbf{Q}_{\text{eff}} = \mathbf{K}_{\zeta\text{-red}}^{(66)} + s\mathbf{E}_{yy} + s^2\mathbf{M}_{yy}$,

$$\det(s\mathbf{I}_{69} - \mathbf{A}_{\text{dyn}}) = \det(\mathbf{R}^{-1})\det(\mathbf{C}^{-1})\det\mathbf{Q}_{\text{eff}}(s). \quad (40)$$

Proof. For a modal solution $\hat{\mathbf{q}}(t) = \mathbf{q}_s \exp(st)$, the first block row of Eq. (39) gives $\hat{\mathbf{v}} = s\hat{\mathbf{u}}$. The remaining rows reduce exactly to Eqs. (36) and (37); Proposition 1 supplies the unique algebraic amplitude. Conversely, a finite pencil eigenvector and $\hat{\mathbf{v}} = s\hat{\mathbf{u}}$ satisfy every generator row. Nonsingularity of $\mathbf{R}$ and $\mathbf{C}$ excludes a hidden differential constraint. $\square$

This result concerns finite roots. The 18 eliminated microfield coordinates remain structural algebraic coordinates in a first-order augmented descriptor; they are not reinterpreted as very fast physical modes.

3.1.4. Nonautonomous finite-time propagator

For fixed $(k, \mathbf{n})$, H1–H3 make $\mathbf{A}_{\text{dyn}}(\cdot; k, \mathbf{n})$ piecewise continuous. The initial-value problem therefore has the unique absolutely continuous solution

$$\hat{\mathbf{q}}(t) = \Phi(t, t_0; k, \mathbf{n})\hat{\mathbf{q}}(t_0), \quad \dot{\Phi} = \mathbf{A}_{\text{dyn}}\Phi, \quad \Phi(t_0, t_0) = \mathbf{I}_{69}. \quad (41)$$

Equivalently,

$$\Phi(t, t_0; k, \mathbf{n}) = \mathcal{T} \exp \left[\int_{t_0}^t \mathbf{A}_{\text{dyn}}(\xi; k, \mathbf{n}) d\xi \right], \quad (42)$$

where $\mathcal{T}$ denotes chronological ordering. Equation (42), rather than an integral of the frozen spectral abscissa, is the primary finite-time object. It

retains mode rotation, branch interaction, and non-normal transient amplification.

At every time, the eliminated field is reconstructed from the simultaneous retained state using Eq. (33). No $\dot{\mathbf{K}}_{\zeta\zeta}$ connection term appears in the 69-state equations because ζ has no differential row or column; such a term would arise only if a time derivative of the reconstructed algebraic observable itself were requested.

3.1.5. Fixed physical scaling and observable gain

Because displacement, velocity, temperature, plastic-chart coordinates, and dislocation densities have different units, an unscaled Euclidean singular value is not a physical invariant. Fix one positive diagonal scale $\mathbf{D}_q$ for the entire history and define

$$\hat{\mathbf{q}} = \mathbf{D}_q \tilde{\mathbf{q}}, \quad \tilde{\mathbf{A}}_{\text{dyn}} = \mathbf{D}_q^{-1} \mathbf{A}_{\text{dyn}} \mathbf{D}_q, \quad \tilde{\mathbf{\Phi}} = \mathbf{D}_q^{-1} \mathbf{\Phi} \mathbf{D}_q. \quad (43)$$

A time-dependent $\mathbf{D}_q$ is not used; it would add $-\mathbf{D}_q^{-1} \dot{\mathbf{D}}_q$ to the scaled generator and would change the observable being measured.

Let $\mathbf{P}_{\text{in}}$ and $\mathbf{P}_{\text{out}}$ be preregistered real coordinate-selection matrices. The pointwise observable gain is

$$\mathcal{G}_D(t; k, \mathbf{n}) = \sigma_1 \left[\mathbf{P}_{\text{out}} \tilde{\mathbf{\Phi}}(t, t_0; k, \mathbf{n}) \mathbf{P}_{\text{in}}^T \right]. \quad (44)$$

The localization observable used here selects the 63 temperature and active crystal-state coordinates at input and output and excludes displacement and velocity from the singular-value objective. Elastic inertia nevertheless remains fully coupled inside $\mathbf{A}_{\text{dyn}}$.

Two gains must be distinguished. The projected quantity $\mathcal{G}_D$ in Eq. (44) answers the conditioned localization input–output question and is used for direction–wavenumber selection. The like-for-like comparison with the spectral abscissa instead uses the complete reduced state,

$$\mathcal{G}_D^{\text{full}}(t; k, \mathbf{n}) = \|\tilde{\mathbf{\Phi}}(t, t_0; k, \mathbf{n})\|_2, \quad \mathbf{P}_{\text{in}} = \mathbf{P}_{\text{out}} = \mathbf{I}_{69}. \quad (45)$$

This separation is necessary because the selected 63-coordinate constitutive subspace is not invariant under the HCP generator: mechanical states can feed the selected output and selected inputs can generate mechanical response. Consequently, a projected gain cannot in general be assigned the full-state frozen spectral abscissa without introducing a controllability/observability mismatch.

Proposition 3 (equivalent norms and conditioned onset)... Let $\mathbf{D}_1$ and $\mathbf{D}_2$ be positive diagonal scales and let the same coordinate selectors be used. Define

$$c_+^{12} = \|\mathbf{P}_{\text{out}}\mathbf{D}_1^{-1}\mathbf{D}_2\mathbf{P}_{\text{out}}^\top\|_2 \|\mathbf{P}_{\text{in}}\mathbf{D}_2^{-1}\mathbf{D}_1\mathbf{P}_{\text{in}}^\top\|_2, \quad (46)$$

$$c_-^{12} = [\|\mathbf{P}_{\text{out}}\mathbf{D}_2^{-1}\mathbf{D}_1\mathbf{P}_{\text{out}}^\top\|_2 \|\mathbf{P}_{\text{in}}\mathbf{D}_1^{-1}\mathbf{D}_2\mathbf{P}_{\text{in}}^\top\|_2]^{-1}. \quad (47)$$

Then

$$c_-^{12}\mathcal{G}_{D_2} \leq \mathcal{G}_{D_1} \leq c_+^{12}\mathcal{G}_{D_2}. \quad (48)$$

Proof. Coordinate selectors commute with the two diagonal scalings on their selected subspaces. The observed matrix in norm 1 is therefore the observed matrix in norm 2 multiplied by the two restricted change-of-scale matrices. Singular value submultiplicativity gives the upper inequality; interchanging 1 and 2 gives the lower inequality. $\square$

Thus finite-time boundedness or blow-up is invariant within an equivalence class of fixed finite-dimensional norms. A numerical threshold-crossing time is not invariant: it must always be reported together with $\mathbf{D}_q$, the selectors, and the threshold.

3.1.6. Continuous direction–wavenumber selection

First choose the compact covering domain

$$\widehat{\Omega}_{nk} = \mathbb{S}^2 \times [k_{\min}, k_{\max}], \quad 0 < k_{\min} < k_{\max} < \infty, \quad (49)$$

and let $\Omega_{nk} = \mathbb{RP}^2 \times [k_{\min}, k_{\max}]$ denote its antipodal quotient. The matrix-valued generator is defined on $\mathbb{S}^2$, not directly on $\mathbb{RP}^2$; the scalar gain descends to the quotient only after the conjugacy in Eq. (52) is established. Define the continuous gain envelope and the finite-time critical time by

$$\Gamma_D(t) = \max_{([\mathbf{n}], k) \in \Omega_{nk}} \mathcal{G}_D(t; k, \mathbf{n}), \quad (50)$$

$$t_c^{\text{FT}}(\mathbf{D}_q, G_c) = \inf\{t \in [t_0, t_f] : \Gamma_D(t) \geq G_c\}, \quad G_c > 1, \quad (51)$$

with $\inf \emptyset = +\infty$. The registered calculation uses $G_c = e$, so onset means one natural-log unit of maximum observable amplification in the locked dimensionless norm; it is not a material constant. The displayed $t_c^{\text{FT}}(\mathbf{D}_q, G_c)$ is shorthand: it also depends on $\mathbf{P}_{\text{in}}$, $\mathbf{P}_{\text{out}}$, t_0 , the base path, material parameters, and Ω_{nk} . No monotonicity of $\Gamma_D(t)$ is assumed.

Proposition 4 (existence and antipodal symmetry)... If H1–H5 hold and the admitted blocks depend continuously on $(\mathbf{n}, k) \in \widehat{\Omega}_{nk}$, then the maximum in Eq. (50) is attained. For real base-state tangents and real selectors/scales,

$$\mathbf{A}_{\text{dyn}}(t; k, -\mathbf{n}) = \overline{\mathbf{A}_{\text{dyn}}(t; k, \mathbf{n})}, \quad \mathcal{G}_D(t; k, -\mathbf{n}) = \mathcal{G}_D(t; k, \mathbf{n}). \quad (52)$$

Proof. Continuous dependence of a linear initial-value problem on its coefficients makes Φ and hence $\mathcal{G}_D$ continuous on $\widehat{\Omega}_{nk}$. Reversing $\mathbf{n}$ changes the sign of every ik directional block and leaves the k^2 blocks unchanged, which complex-conjugates the real-coefficient generator and its propagator. Complex-conjugate matrices have the same singular values. Consequently, the continuous scalar gain descends to the compact quotient Ω_{nk} , where the Weierstrass theorem gives an attained maximum. $\square$

Uniqueness is not presumed. For a relative gain tolerance $0 < \varepsilon_G < 1$, the robust selector is the near-optimal set

$$\mathcal{S}_\varepsilon(t) = \{([\mathbf{n}], k) \in \Omega_{nk} : \mathcal{G}_D(t; k, \mathbf{n}) \geq (1 - \varepsilon_G)\Gamma_D(t)\}. \quad (53)$$

For fixed nonzero ε_G , this set generally has nonzero extent even around a smooth strict maximum. Numerical resolution requires Hausdorff stability of $\mathcal{S}_\varepsilon$ and convergence of the maximizer candidates under time-step, angular, radial, and search-box refinement. A unique $(\mathbf{n}_{\text{FT}}^*, k_{\text{FT}}^*)$ is reported only when the limiting maximum is strict and isolated and the conclusion persists as $\varepsilon_G \rightarrow 0$; otherwise the selector is explicitly set-valued. The associated linear wavelength is $\lambda_{\text{FT}}^* = 2\pi/k_{\text{FT}}^*$ only for an interior, refinement-stable wavenumber.

3.1.7. Long-wave and high-wavenumber robustness

For $k > 0$, introduce

$$\widehat{\mathbf{q}}_w = [\widehat{\mathbf{w}}, \dot{\widehat{\mathbf{w}}}, \widehat{\mathbf{d}}]^\top = \mathbf{T}_k \widehat{\mathbf{q}}, \quad \mathbf{T}_k = \text{diag}(ik\mathbf{I}_3, ik\mathbf{I}_3, \mathbf{I}_{63}), \quad (54)$$

so that $\mathbf{A}_w = \mathbf{T}_k \mathbf{A}_{\text{dyn}} \mathbf{T}_k^{-1}$. If, in addition to H1–H3, $\mathbf{K}_{\zeta\zeta}^{-1}$ remains uniform at $k = 0$ and the reduced blocks have continuous leading limits with

$$\mathbf{K}_{uu} = O(k^2), \quad \mathbf{K}_{ud} = O(k), \quad \mathbf{K}_{du} = O(k), \quad \mathbf{K}_{dd} = O(1), \quad (55)$$

then $\mathbf{A}_w$ has a finite $k \rightarrow 0^+$ limit after rigid translation is projected out. Because the registered input and output selectors act only on $\mathbf{d}$, their observed propagator block is unchanged by $\mathbf{T}_k$; the observable gain therefore

has the same finite conditional limit. The present implementation propagates $[\widehat{\mathbf{u}}, \widehat{\mathbf{v}}, \widehat{\mathbf{d}}]$ only for $k > 0$ and its primary search begins at a positive $k_{\min}$; it does not numerically certify the hypotheses in Eq. (55). Consequently, a maximum that follows $k_{\min}$ under domain enlargement is reported only as a long-wave boundary diagnostic, not as a finite material wavelength or a proved $k = 0$ continuation.

At high wavenumber, positivity of the acoustic tensor, directional conductivity, and the relevant gradient principal symbol is a necessary admission test, but by itself is not a proof against non-normal transient growth. The following conditional statement supplies the required continuous bound.

Proposition 5 (conditional uniform high- k observable bound). Suppose that for $k \geq k_H$, all $\mathbf{n} \in \mathbb{S}^2$, and $t \in I$ there is a Hermitian, differentiable symmetrizer for the fixed-scale generator $\widetilde{\mathbf{A}}_{\text{dyn}}$, $\mathbf{W}(t; k, \mathbf{n}) > \mathbf{0}$ such that

$$\dot{\mathbf{W}} + \mathbf{W} \widetilde{\mathbf{A}}_{\text{dyn}} + \widetilde{\mathbf{A}}_{\text{dyn}}^H \mathbf{W} \preceq 2\omega_\infty(t) \mathbf{W}, \quad \omega_\infty \in L^1(I), \quad (56)$$

with ω_∞ independent of k and $\mathbf{n}$. Suppose also that the selector-energy constants

$$C_{\text{in}} = \sup_{\substack{k \geq k_H \\ \mathbf{n} \in \mathbb{S}^2}} \|\mathbf{W}(t_0; k, \mathbf{n})^{1/2} \mathbf{P}_{\text{in}}^T\|_2 < \infty, \quad (57)$$

$$C_{\text{out}} = \sup_{\substack{t \in I, k \geq k_H \\ \mathbf{n} \in \mathbb{S}^2}} \|\mathbf{P}_{\text{out}} \mathbf{W}(t; k, \mathbf{n})^{-1/2}\|_2 < \infty \quad (58)$$

are finite. Then

$$\sup_{\substack{k \geq k_H \\ \mathbf{n} \in \mathbb{S}^2}} \mathcal{G}_D(t; k, \mathbf{n}) \leq C_{\text{out}} C_{\text{in}} \exp\left(\int_{t_0}^t \omega_\infty(\xi) d\xi\right). \quad (59)$$

Proof. For $V = \widetilde{\mathbf{q}}^H \mathbf{W} \widetilde{\mathbf{q}}$, Eq. (56) gives $\dot{V} \leq 2\omega_\infty V$. The input selector enters this energy norm with factor C_{in} , Gronwall propagates it, and the output selector returns to the Euclidean observation norm with factor C_{out} , yielding Eq. (59). $\square$

Equation (59) proves uniform finite-time boundedness of the declared observable under its stated symmetrizer and selector hypotheses; it does not prove full-state boundedness or asymptotic decay. A stronger uniform equivalence $m\mathbf{I} \preceq \mathbf{W} \preceq M\mathbf{I}$ would bound the full scaled propagator by

$\sqrt{M/m}$ times the same exponential, but standard acoustic displacement–velocity coordinates need not admit such a k -uniform equivalence. In the present workflow, positive principal-symbol margins, algebraic conditioning, modal residuals, and extension of the numerical scan are reported only as consistency diagnostics along the audited paths. They do not construct $\mathbf{W}$, do not verify the hypothesis on a continuum, and are not promoted to an unconditional theorem for all k .

3.1.8. Dominant mechanism and nonlinear-width bridge

Let $\tilde{\mathbf{q}}_{\text{out}}^*$ be a unit left singular vector of the maximizing observed propagator, embedded in the 69-coordinate scaled state. For the mutually disjoint blocks $b \in \{\text{thermal, plastic, dislocation, mechanical}\}$, define

$$\eta_b^{\text{FT,out}} = \frac{\|\mathbf{P}_b \tilde{\mathbf{q}}_{\text{out}}^*\|_2^2}{\sum_c \|\mathbf{P}_c \tilde{\mathbf{q}}_{\text{out}}^*\|_2^2}, \quad \sum_b \eta_b^{\text{FT,out}} = 1. \quad (60)$$

Here the plastic block contains the plastic-distortion chart and signed slip; the dislocation block contains mobile and dipole densities. A zero mechanical fraction follows automatically when mechanical coordinates are excluded by the output selector and does not imply that inertia was removed from the dynamics. The algebraic microfield may be reconstructed and reported separately. To make “nearly repeated” operational before inspecting a mechanism vector, we define

$$\delta_\sigma = \frac{\sigma_1 - \sigma_2}{\sigma_1}, \quad \delta_\sigma < 0.05 \quad \Longleftrightarrow \quad \frac{\sigma_1}{\sigma_2} < 1.0526. \quad (61)$$

When this gate is triggered, only the range of the leading singular subspace is interpreted; fractions from one arbitrary basis vector are suppressed. For a rank-one observed map, such as a scalar temperature output, σ_2 does not exist and the ratio is reported as not applicable rather than as evidence of separation. Onset and terminal ratios are reported for every selector in the reproducibility table. When $\mathcal{S}_\varepsilon$ contains distinct direction–wavenumber branches, Eq. (60) is evaluated for every branch and reported branchwise or as a range; one winning branch is not used to represent the whole near-optimal set. The $\eta_b^{\text{FT,out}}$ are locked-scale coordinate fractions of an optimal singular vector. They are not energy fractions, dissipation fractions, or identified causal contributions.

For nonlinear and linear windows $I_{\text{NL}} = [t_{0,\text{NL}}, t_{\text{NL}}]$ and $I_{\text{L}} = [t_{0,\text{L}}, t_{\text{L}}]$, the downstream computational bridge is written

$$W_{b,\text{FWHM}}^{(c)}(t_{\text{NL}}) = C_{\text{FWHM}}^{(c,b)}(I_{\text{NL}}; I_{\text{L}}) \frac{2\pi}{k_{\text{FT}}^*(t_{\text{L}})}, \quad (62)$$

where c identifies the perturbation class and b the temperature or plastic shear-rate observable. The coefficient $C_{\text{FWHM}}^{(c,b)}$ is accepted only after nested-grid convergence with one fixed grain map. An onset-wavelength ratio is retained only as a memory diagnostic, and a set-valued k_{FT}^* induces a reported interval rather than a single coefficient. A same-window quantitative relation requires both $t_{0,\text{L}} = t_{0,\text{NL}}$ and $t_{\text{L}} = t_{\text{NL}}$. Matching only the terminal time is labelled same-terminal-time and remains conditioned on unequal history windows; unequal terminal times define a cross-horizon diagnostic. Neither is silently relabeled as a same-window calibration. The planned classical benchmark uses constant top velocity, a linear initial velocity field, uniform initial temperature, and separate unperturbed, broadband-perturbed, and orientation-heterogeneous cases. Equation (62) is a computational relation, not an experimental parameter identification or an independent full-field validation claim.

4. Evidence design and numerical implementation

4.1. Base history, exact generator and propagation

The repository verification seed uses the complete 18-slip closure in Section 2, a slip-gradient length $\ell_{\chi} = 0.25 \mu\text{m}$ and a Nye length $\ell_N = 1 \mu\text{m}$. Its grouped constraints are informed by titanium literature, but the card is not identified for a particular batch or specimen. The base history is integrated independently and then stored at 129 accepted states on $[0.5, 140] \mu\text{s}$. At each state, the continuous constitutive Jacobian in Eq. (16) is assembled on the recorded smooth branch. A state is rejected if a derivative probe changes that branch or if

$$\text{cond}_2(\mathbf{K}_{\zeta\zeta}) \geq 10^{14}, \quad \text{cond}_2(\mathbf{R}) \geq 10^{12}, \quad \text{or} \quad \text{cond}_2(\mathbf{C}) \geq 10^{12}. \quad (63)$$

The algebraic microfield is eliminated, the generator in Eq. (39) is assembled and the block reconstruction residual is recorded.

Between stored states t_i and t_{i+1} , the fixed-scale generator is interpolated linearly. Four exponential-midpoint substeps are used per interval:

$$\widehat{\mathbf{A}}_{ij} = (1 - f_j)\widehat{\mathbf{A}}_i + f_j\widehat{\mathbf{A}}_{i+1}, \quad f_j = (j + 1/2)/4, \quad (64)$$

$$\Phi_{ij+1} = \exp(h_i\widehat{\mathbf{A}}_{ij})\Phi_{ij}, \quad h_i = (t_{i+1} - t_i)/4. \quad (65)$$

The reference history therefore uses 512 ordered exponentials. Base-history refinement and propagation-substep refinement are audited separately.

The diagonal coordinate scale $\mathbf{D}_q$ is fixed before any spectrum or singular vector is inspected. Representative scales are 1 μm for displacement, 0.1 ms^{-1} for velocity, 1 K for temperature, 2×10^{-4} for plastic-chart and signed-slip coordinates, $1.6 \times 10^{12} \text{ m}^{-2}$ for mobile density and 10^{10} m^{-2} for dipole density. Input and output selectors retain temperature and the 62 local coordinates. Displacement and velocity remain coupled inside the propagator but are not directly optimized. The declared threshold is $\mathcal{G}_c = e$.

4.2. Independent descriptor audit and classical anchor

The quadratic pencil is independently converted to an 87-coordinate first-order generalized eigenproblem with state ordering $[\widehat{\mathbf{u}}, \widehat{\mathbf{v}}, \widehat{T}, \widehat{\boldsymbol{\zeta}}, \widehat{\mathbf{a}}]$. Generalized Schur decomposition must return 69 finite roots and 18 infinite roots. The finite roots are matched to eigenvalues of the reduced 69-state generator, and both modal backward error and microfield reconstruction residual are evaluated. No small mass or relaxation time is assigned to the algebraic coordinates.

Separately, the scalar Bai coefficient/root implementation is checked against exact-dyadic sign and zero tests and targeted regression cases. This establishes an executable classical anchor. It is not used as evidence of a strict analytical HCP-to-Bai reduction; that reduction remains a separate open gate.

4.3. Central finite-time/frozen-spectrum discrimination

For each fixed branch $(k, \mathbf{n})$, define the frozen spectral abscissa of the same scaled 69-state generator as

$$\alpha(t; k, \mathbf{n}) = \max_j \text{Re } \lambda_j[\widetilde{\mathbf{A}}_{\text{dyn}}(t; k, \mathbf{n})] \quad (66)$$

and its accumulated prediction

$$\mathcal{I}_\alpha(t; k, \mathbf{n}) = \int_{t_0}^t \alpha(\xi; k, \mathbf{n}) \, d\xi. \quad (67)$$

The primary like-for-like finite-time quantity is the full-state log gain

$$\mathcal{L}_{\text{FT}}^{\text{full}}(t; k, \mathbf{n}) = \log \|\tilde{\Phi}(t, t_0; k, \mathbf{n})\|_2, \quad (68)$$

and the discrimination statistic is

$$\Delta_{\text{FT}}^{\text{full}} = \mathcal{L}_{\text{FT}}^{\text{full}} - \mathcal{I}_{\alpha}. \quad (69)$$

Equations (66)–(69) use the same complete 69-state space, fixed scaling and identity input/output selectors. The projected localization gain in Eq. (44) is reported separately and is not subtracted from the full-state α . The selected 63-coordinate subspace has nonzero mechanical input and output coupling, so it is not a closed instantaneous system.

As a same-norm upper-bound audit, define the numerical abscissa

$$\mu_2(\tilde{\mathbf{A}}) = \lambda_{\max} \left[\frac{\tilde{\mathbf{A}} + \tilde{\mathbf{A}}^{\text{H}}}{2} \right], \quad \log \|\tilde{\Phi}(t, t_0)\|_2 \leq \int_{t_0}^t \mu_2(\tilde{\mathbf{A}}) dt. \quad (70)$$

The bound is not used as a selector; it checks that the propagated full-state gain is consistent with the declared norm.

To verify both the implementation and the sufficiency statement, let

$$\tilde{\mathbf{A}}(t) = \mathbf{U} \text{diag}[\lambda_1(t), \dots, \lambda_m(t)] \mathbf{U}^{\text{H}}, \quad \mathbf{U}^{\text{H}} \mathbf{U} = \mathbf{I}, \quad (71)$$

where the basis is independent of time. The family is normal and pairwise commuting, and hence

$$\log \|\Phi(t, t_0)\|_2 = \max_j \int_{t_0}^t \text{Re } \lambda_j(\xi) d\xi. \quad (72)$$

This equals $\int \alpha dt$ if one common mode attains the pointwise maximum almost everywhere. A stationary normal control and a time-varying commuting-normal control with a fixed leader test this equality using the same exponential-midpoint/frozen-integral quadrature. A third commuting control switches its pointwise leader and must fail the equality; it demonstrates that commutation alone is insufficient. The HCP test is then applied to the optimized near-onset x - and y -like branches and to the terminal y -like branch.

A second stationary diagonal control applies two invariant coordinate projections. When the frozen rate is restricted to the same invariant subspace,

projected gain and frozen accumulation coincide; when the selected subspace excludes the full-state leader, comparison with the full-state α fails by construction. This control detects precisely the input–output mismatch avoided in Eq. (69).

The source of HCP non-equivalence is diagnosed with four quantities. For adjacent scaled generators, the normalized commutator is

$$c_i = \frac{\|\tilde{\mathbf{A}}_{i+1}\tilde{\mathbf{A}}_i - \tilde{\mathbf{A}}_i\tilde{\mathbf{A}}_{i+1}\|_F}{\|\tilde{\mathbf{A}}_{i+1}\|_F\|\tilde{\mathbf{A}}_i\|_F}. \quad (73)$$

For the instantaneous leading eigenvalue, left and right eigenvectors give

$$\kappa_\lambda = \frac{\|\mathbf{w}\|_2\|\mathbf{v}\|_2}{|\mathbf{w}^H\mathbf{v}|}, \quad \theta_i = \cos^{-1} \frac{|\mathbf{v}_i^H\mathbf{v}_{i+1}|}{\|\mathbf{v}_i\|_2\|\mathbf{v}_{i+1}\|_2}. \quad (74)$$

Finally, the chronological product is compared with the reverse-ordered product and with $\exp(\int \mathbf{A} dt)$, the first Magnus surrogate. These tests separate mode ill-conditioning, eigenvector rotation and operator ordering from the mere observation that two terminal numbers differ.

4.4. Cross-orientation transfer test

Three Bunge orientations, $(0^\circ, 0^\circ, 0^\circ)$, $(30^\circ, 90^\circ, 30^\circ)$ and $(90^\circ, 60^\circ, 30^\circ)$, are selected independently as the minimum-, median- and maximum-storage representatives of the pre-existing deterministic orientation atlas. For each orientation, its own direction–wavenumber pair is refined; the full-state propagator norm and frozen integral are then recomputed on that same pair, base path and scaling. No orientation is fitted to increase $|\Delta_{\text{FT}}^{\text{full}}|$. The transfer gate requires at least three orientations, modal backward errors below 10^{-8} and $|\Delta_{\text{FT}}^{\text{full}}| \geq 0.1$ at either the selection or terminal horizon for every orientation.

4.5. Direction–wavenumber search and horizon tracking

The primary search covers $300 \leq k \leq 3 \times 10^5 \text{ m}^{-1}$ and one closed hemisphere, using the antipodal equivalence $\mathbf{n} \sim -\mathbf{n}$. A nested scrambled Sobol sequence samples the projective sphere and $\log k$ at 64, 256 and 1024 points. Exact sample axes and the independently registered onset/final branches are retained as anchors because a narrow maximum on the equator can be missed by an unanchored finite spherical design. Empirical projective-sphere fill angle and maximum $\log k$ gap are measured at every level. Eight mutually separated starts are refined independently for both the onset and terminal

objectives, with every function evaluation retained. Candidates are retained only if the last two anchor-combined maxima differ by less than 1%, at least two starts recur in the winning basin, k^* remains strictly inside the registered interval, and reevaluation on the 129-state/four-substep history lies within 1% in gain of the independently registered reference candidate. This is an *anchor-assisted basin-retention and convergence audit*. It is not called a global-search certificate because the blind Sobol design fails to discover the narrow terminal basin. At each horizon, branches within

$$\mathcal{G}(t; k, \mathbf{n}) \geq (1 - \varepsilon_G) \Gamma_D(t), \quad \varepsilon_G = 0.02, \quad (75)$$

are retained as a near-optimal set. Temporal convergence is evaluated on 65- and 129-state histories with one, two, four and eight midpoint substeps. Norm sensitivity uses the baseline scale plus independent factor-two increases and decreases of mechanical, thermal, dislocation, plastic-kinematic and all-constitutive groups, giving 11 fixed norms. Seven selector pairs test full/full, constitutive/constitutive, full/constitutive, constitutive/full, mechanical/mechanical, mechanical/constitutive and constitutive/temperature maps. The blind Sobol maximum and the anchor-combined maximum are reported separately so that anchor dependence cannot be hidden by the audit label.

The original 11-by-7 table ranks the three registered branch candidates and is retained as a complete fixed-candidate screen in the Supplementary Material. To test whether that restriction hides a selector-specific basin, all seven selectors are additionally re-optimized at onset and terminal time under the baseline norm and four adverse norms selected before re-optimization: mechanical $\times 0.5$, thermal $\times 0.5$, dislocation $\times 2$ and all-constitutive $\times 2$. A shared 128-point scrambled Sobol design, exact axes and the three registered branches seed the bounded local searches; every objective is then evaluated on the 129-state/four-substep reference. A compact-domain boundary maximum is reported as such. In particular, full/full and mechanical/mechanical maps can select reversible high-frequency elastic waves and are not required to define an interior localization wavelength. Selectors that observe a constitutive or thermal localization response must retain an interior k^* .

For a strict same-input/output freezing baseline, divide every registered nonuniform interval into r equal subintervals and define

$$\Phi_{\text{PF}}^{(r)} = \prod_{m=N_r}^1 \exp \left[\Delta t_m \tilde{\mathbf{A}}(t_{m-1}) \right], \quad \mathcal{G}_{\text{PF}}^{(r)} = \left\| \mathbf{P}_{\text{out}} \Phi_{\text{PF}}^{(r)} \mathbf{P}_{\text{in}}^T \right\|_2. \quad (76)$$

The metric and both selectors are identical to those of the time-ordered reference; only the generator is frozen at the left endpoint of each subinterval. Results are reported for $r = 1, 2, 4, 8, 16$ subdivisions per registered interval. This is a controlled discretization of chronology, not the incompatible scalar $\int \alpha dt$ comparison.

4.6. Loading transfer, local parameter sensitivity and external titanium check

Three additional paths are integrated without refitting: simple shear at 5.0×10^3 and $1.3 \times 10^4 \text{ s}^{-1}$ from 293.15 K, and the baseline $1.0 \times 10^4 \text{ s}^{-1}$ path from 373.15 K. Direction and wavenumber are re-optimized for the constitutive/constitutive map at onset and terminal time using shared Sobol reconnaissance plus continuation from the baseline basin. These rate/temperature cases test transfer, not specimen prediction.

Local sensitivity uses symmetric multiplicative perturbations of $\pm 20\%$ for the initial mobile density ρ_{m0} , mean-free-path coefficient K_Λ , Nye length ℓ_N and full-rank slip-gradient length ℓ_χ . The same branch is continued from the baseline and direction–wavenumber optimization is repeated. Central logarithmic sensitivities of gain and wavenumber are

$$S_p^Y = \frac{\log Y(1.2p) - \log Y(0.8p)}{\log(1.2) - \log(0.8)}, \quad Y \in \{\mathcal{G}, k^*\}. \quad (77)$$

This is a branch-local derivative audit, not global uncertainty quantification.

Finally, the $1.3 \times 10^4 \text{ s}^{-1}$ calculation is compared with the synchronized shear-stress and infrared-temperature landmarks measured for commercial-purity Grade-II titanium by Guo et al. [20]. Peak and shear-band times are read from their Fig. 1; the temperature-rise interval at peak stress and localized-temperature interval are taken from the paper text. Because specimen geometry, texture, damage and localization are not matched, this is an external timing/scale check with explicit failures, not parameter identification.

4.7. Fixed-base coupling interventions

To distinguish coordinate participation from dependence on constitutive coupling, the 69-state generator is partitioned into mechanical/inertial, thermal, dislocation and plastic-kinematic groups. Within-group blocks and the base history are held fixed. Six cross-group edge families are switched on or off, and the leading singular value and optimal vectors are recomputed

for all $2^6 = 64$ coalitions. The value function is the maximum finite-time observable log gain. Exact Shapley allocation is used to summarize edge dependence, while grouped removals (no inertial, no thermal or no dislocation cross-links) provide transparent counterfactual checks. These operations identify dependence of the *linearized fixed-base model* on selected blocks; they are not experimental causal effects, dissipation fractions or unique structural mechanisms.

Gradient controls are reported in the same figure but under a separate contract. They use an assembled frozen quasi-static HCP operator, extend the sampled range to $k = 10^{12} \text{ m}^{-1}$, and test removal of slip-gradient and Nye contributions. They do not supply nonlinear widths or a calibrated high- k theorem, so no quantitative comparison between the dynamic coupling-intervention panel and the quasi-static gradient panel is made.

4.8. Singular-vector nonlinear transport and material-width gates

The first nonlinear gate uses the actual leading right singular vectors of the re-optimized baseline constitutive/constitutive map at $1.4375 \mu\text{s}$ and $140 \mu\text{s}$. Candidate direction and wavenumber and the coordinate metric are held at their released V3 values. The fixed- β , passive-ledger base path is independently refined from 129 to 513 states, and the singular vectors are recomputed on that refined map before nonlinear transport. Dense-history convergence compares nested 257- and 513-state maps with four midpoint substeps, followed by a one-to-four-substep check on the 513-state map. If the 257-to-513 relative log-gain change exceeds 2%, a 1025-state map is mandatory and convergence is judged from the 513-to-1025 change under the same four-substep contract. The released 129-state gain is retained as a coarse-history screen; a finer value supersedes it only after this density gate closes, without changing the registered direction, wavenumber or metric. Each vector is imposed as a real periodic Fourier mode with 16 cells per wavelength and two factor-two amplitudes. The first replay retains non-negative modes 0–3 and is therefore labelled a Galerkin transport check, not a complete Fourier calculation. A separate audit retains every nonnegative mode 0–8 resolvable on the 16-cell grid and repeats the replay with one and four nonlinear midpoint substeps. Any non-finite continuation or more than 1% discarded right-hand-side Fourier energy leaves the complete nonlinear gate open. The terminal amplitude is additionally limited so that its linearly predicted final dimensionless norm does not exceed 10^{-3} . The nonlinear periodic momentum–heat–micromorphic–crystal-state residual is advanced with

the registered tangent used only as an integrating factor. Acceptance requires less than 2% error in gain, less than 3% error in the complete complex Fourier output vector, less than 2% factor-two gain variation, less than 1% nonretained Fourier energy within the declared subspace and a power-partition ledger residual below 10^{-10} . Passing these Galerkin checks does not override a failed full-Fourier audit.

A separate 0–2 μs material-width bridge imposes prescribed sinusoidal or multimode disturbances with locked phases and spatial grids. It is accepted in stages. First, linear gain and phase transport must be reproduced. Second, multiple input amplitudes must remain in the amplitude-linear regime for a single mode. Third, the peak and full width at half maximum (FWHM) must converge on nested grids. Finally, a candidate material width must be insensitive to seed phase/spectrum, remain resolved by at least 12 cells and emerge independently of the imposed wavelength. A failure retains the successful lower-level verification but disables the material-scale interpretation.

5. Results

5.1. *The finite-spectrum reduction is structurally exact*

Figure 2 separates three statements that should not be conflated. The scalar Bai implementation passes 34 targeted and 358 related regression tests, including exact-dyadic sign and zero checks, and is therefore a controlled classical reference. The augmented HCP first-order descriptor has dimension 87 and rank 69 in its differential matrix. Generalized Schur decomposition returns exactly 69 finite and 18 infinite roots; the latter correspond one-for-one to the algebraic micromorphic coordinates. Complete matching of the finite descriptor roots to the reduced generator gives a maximum relative error of 8.595×10^{-8} and a median error of 1.01×10^{-22} . The code-level generator and decoupling residuals satisfy 2×10^{-13} gates. Thus exact elimination neither creates a fast microfield mode nor deletes a finite crystal mode.

These checks do not establish a strict analytical limit from the 18-slip HCP operator to Bai’s scalar coefficients. Figure 2 therefore displays that reduction as open rather than treating numerical agreement within each separate model as a proof of cross-model equivalence.

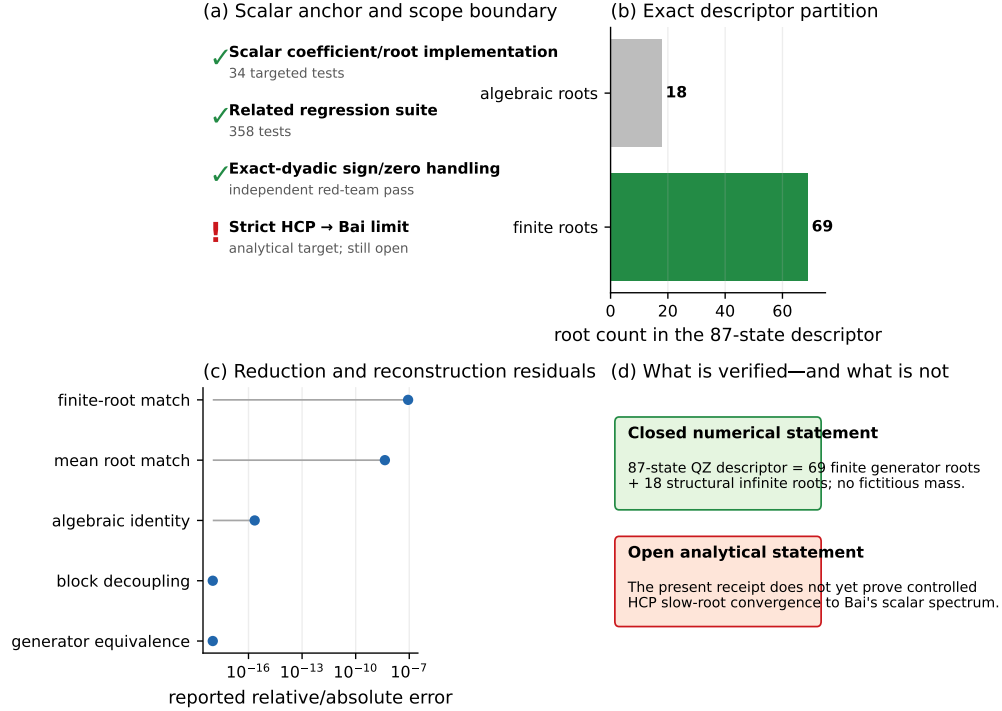


Figure 2: Classical anchor and exact-reduction audit. The Bai scalar implementation passes its scoped regression tests. For the HCP operator, the 87-coordinate augmented descriptor contains 69 finite roots and 18 structural infinite roots; exact algebraic elimination and the 69-state generator reproduce the finite spectrum and reconstruct the eliminated microfield. A strict HCP-to-Bai analytical reduction is deliberately retained as an open gate.

5.2. The base path evolves through distinct constitutive regimes

The homogeneous path is far from a stationary linearization (Fig. 3). First Piola shear stress rises from 176.0 MPa at $0.5 \mu\text{s}$ to a peak of 565.7 MPa at $2 \mu\text{s}$, then decreases to 431.8 MPa at $140 \mu\text{s}$. Over the same interval, temperature rises from 293.15 to 548.83 K and accumulated absolute slip reaches 1.388. Total mobile density grows from 2.88×10^{13} to $2.04 \times 10^{14} \text{ m}^{-2}$, while total dipole density grows from 1.80×10^{11} to $1.95 \times 10^{14} \text{ m}^{-2}$. Prismatic $\langle a \rangle$ slip remains the dominant family, but the resistance, rate, thermal and density tangent blocks change substantially. The accumulated crystal-plastic power ledger is $6.76 \times 10^8 \text{ J m}^{-3}$ at the terminal state, partitioned into $6.08 \times 10^8 \text{ J m}^{-3}$ generated heat and a $6.76 \times 10^7 \text{ J m}^{-3}$ passive nonthermal

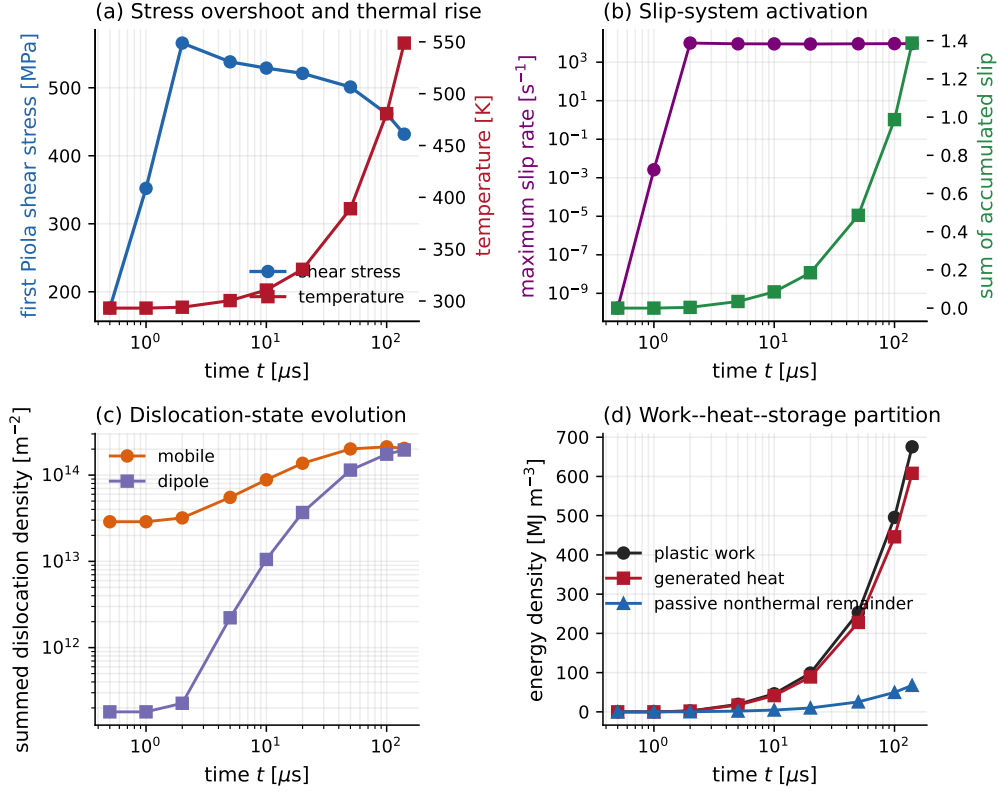


Figure 3: Evolution of the registered homogeneous HCP base state. Stress, temperature, accumulated slip, mobile and dipole dislocation densities, and the plastic-work ledger evolve over the same loading history used to assemble the perturbation generator. The stress peak near $2 \mu\text{s}$ separates early hardening from the later thermally softened path.

remainder. The latter is not interpreted as an identified dislocation free energy. This evolving base path supplies the nonautonomous operator tested below.

5.3. Frozen spectral accumulation is not the finite-time selector

Figure 4 gives the decisive same-branch comparison. At each checkpoint, $\mathcal{I}_\alpha$ and $\mathcal{L}_{\text{FT}}^{\text{full}}$ use the same 129-state base history, direction, wavenumber, scaling, complete 69-state space and identity input/output selectors. The projected constitutive gain is shown only as a conditioned localization diagnostic. The maximum modal backward error in the frozen eigenproblems is 5.33×10^{-24} , so the discrepancy is not attributable to unresolved eigenpairs.

Table 2: Like-for-like full-state finite-time/frozen-spectrum discrimination at 140 μ s. Here $\Delta_{\text{FT}}^{\text{full}} = \log \|\tilde{\Phi}\|_2 - \int \alpha dt$.

Branch	k (m^{-1})	$\log \mathcal{G}$	$\int \alpha dt$	Δ_{FT}
Onset x	24163.9	6.874	2.090	+4.784
Onset y	51247.2	9.939	2.722	+7.217
Terminal y	7724.0	13.368	6.966	+6.402

The three branch results are summarized in Table 2. Frozen accumulation underpredicts the full-state terminal log gain by 4.784, 7.217 and 6.402 on the onset- x , onset- y and terminal- y branches. The originally projected onset- x gain is only 0.902, compared with the full-state value 6.874; this large difference demonstrates why a projected gain cannot be subtracted from the full-state spectral abscissa. For the two y -like branches, full and projected log gains differ by less than 4.3×10^{-4} . No branch-independent additive correction to the frozen growth rate recovers all three trajectories.

The mechanism diagnostics make the discrepancy interpretable. The maximum instantaneous leading-mode condition numbers are 2.96×10^5 , 2.78×10^5 and 6.06×10^2 for the three branches, while maximum adjacent leading-vector rotations are 51.1° , 25.0° and 9.46° . The normalized adjacent commutator is nonzero on every branch. For the terminal branch, reversing the one-step chronological factors reduces the full-state log gain from 13.370 to 10.979, and the first-Magnus surrogate $\exp(\int \tilde{\mathbf{A}} dt)$ gives 10.115. Thus modal ill-conditioning and chronological composition are quantitatively active, not post hoc labels attached to $\Delta_{\text{FT}}^{\text{full}}$.

In the complete 11-by-7 *fixed-candidate* screen, the near-onset winner remains onset- y for full/full, constitutive/constitutive, full/constitutive, mechanical/mechanical and mechanical/constitutive. Constitutive/full and constitutive/temperature instead select the branch labelled terminal- y even at the near-onset horizon. At the terminal horizon, terminal- y wins every tested selector except mechanical/mechanical, which selects onset- x . The three registered branches is therefore robust to the norm perturbations, while the complete matrix demonstrates that changing what is allowed at input or measured at output can change the scientific answer. Because this screen does not optimize outside those three branches, its complete matrix is placed in the Supplementary Material and the unrestricted selector-specific re-optimization is reported below.

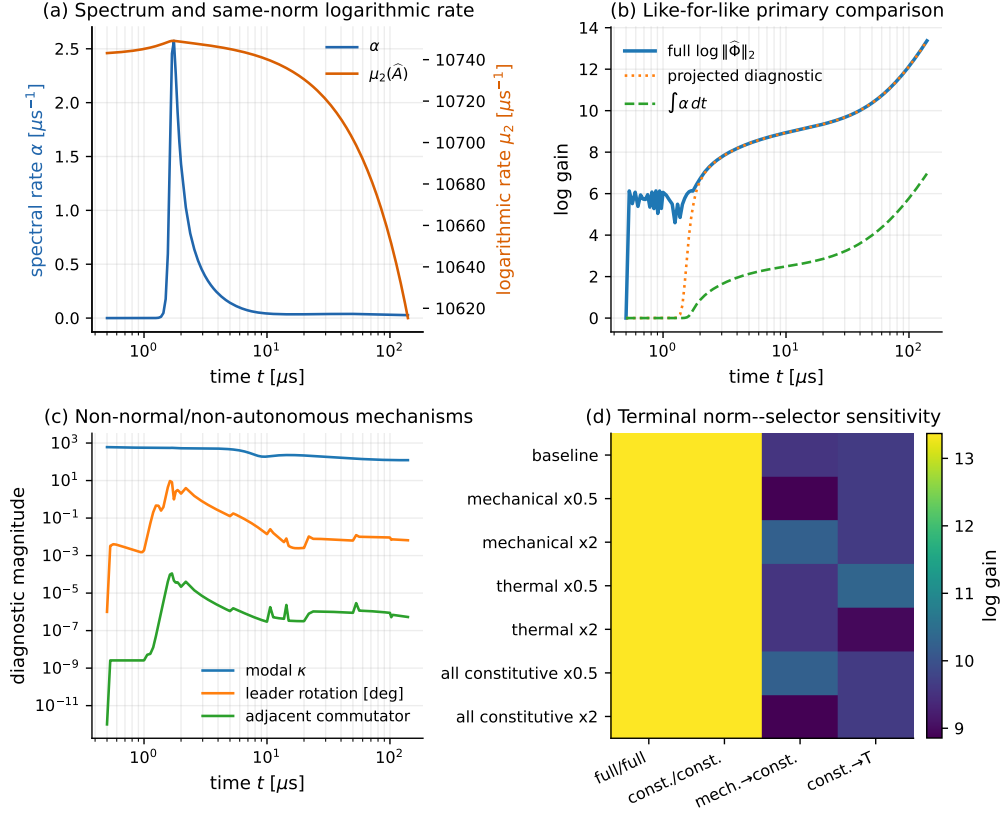


Figure 4: Like-for-like operator and sensitivity audit. (a) Full-state spectral and logarithmic rates for the terminal branch; the latter is a valid but loose same-norm upper bound. (b) Full-state propagator gain and frozen integral on the same 69-state space; the projected constitutive gain is shown separately. (c) Instantaneous leading-mode condition number, adjacent eigenvector rotation and normalized commutator. (d) Terminal gains under representative norm-selector changes; the complete 11-by-7 fixed-candidate matrix is supplied as Supplementary Fig. S1.

5.4. *Positive controls, orientation transfer and anchor-assisted search audit*

The two positive controls in Fig. 5(a) close an important logical loop. For the stationary normal generator, the final difference between the propagated log norm and the frozen integral is 4.44×10^{-15} ; for the time-varying commuting-normal family with one common leading mode it is zero to the reported precision, and the maximum history error is 1.78×10^{-15} . Normality and adjacent-commutator residuals remain below 2.2×10^{-16} . The switching-leader family is also commuting and normal but gives $\Delta_{\text{FT}} = -1.146$, as predicted by the distinction between $\max_j \int \text{Re } \lambda_j dt$ and $\int \max_j \text{Re } \lambda_j dt$. Thus the code recovers equality when its sufficient conditions hold and rejects it when the missing fixed-leader condition is deliberately removed.

The projected stationary control closes the input–output logic separately. Both invariant projections reproduce the frozen integral restricted to the same subspace to within 2×10^{-12} . When the projection excludes the full-state leader, comparison with the unrestricted α intentionally fails. The HCP constitutive subspace has nonzero bidirectional mechanical coupling and is therefore not assigned a projected frozen spectrum.

The HCP non-equivalence transfers to all three independently selected orientations (Fig. 5b). At the selection horizon, Δ_{FT} is 6.234, 5.810 and 5.532 for the minimum-, median- and maximum-storage representatives. At $140 \mu\text{s}$, the corresponding values are 6.737, 4.461 and 3.307. Their maximum frozen-mode backward errors are below 10^{-23} . The magnitude changes with orientation and with the independently selected $(\mathbf{n}, k)$ pair, but non-equivalence does not disappear. The central result is therefore not a peculiarity of the $(0^\circ, 0^\circ, 0^\circ)$ path.

The anchor-assisted audit challenges retention of the selector without claiming independent global discovery (Fig. 5c,d). From 64 to 1024 Sobol samples, the empirical projective-sphere fill angle decreases from 18.52° to 4.45° and the maximum $\log k$ gap from 0.199 to 0.0116. The anchor-combined onset and terminal maxima are unchanged over the last two levels. Four of eight onset refinements and five of eight terminal refinements return to the retained winning basin. On the 129-state/four-substep history, the new onset candidate has gain 5.32086 compared with the independently registered 5.32074, while the terminal candidate has gain 6.355×10^5 , only 0.516% below the reported 6.388×10^5 candidate. Both wavenumbers remain in the interior of the search interval.

The blind Sobol result is retained as a failed discovery test: at the terminal horizon its largest log gain rises only from 2.20 to 4.18, far below

the anchor-combined 13.35. The terminal basin is too narrow for 1024 unanchored points. Accordingly, the accepted statement is only that exact symmetry/prior-branch anchors, independent basin recurrence and high-resolution verification retain the reported basin on the registered domain. The calculation is not a global-optimum certificate.

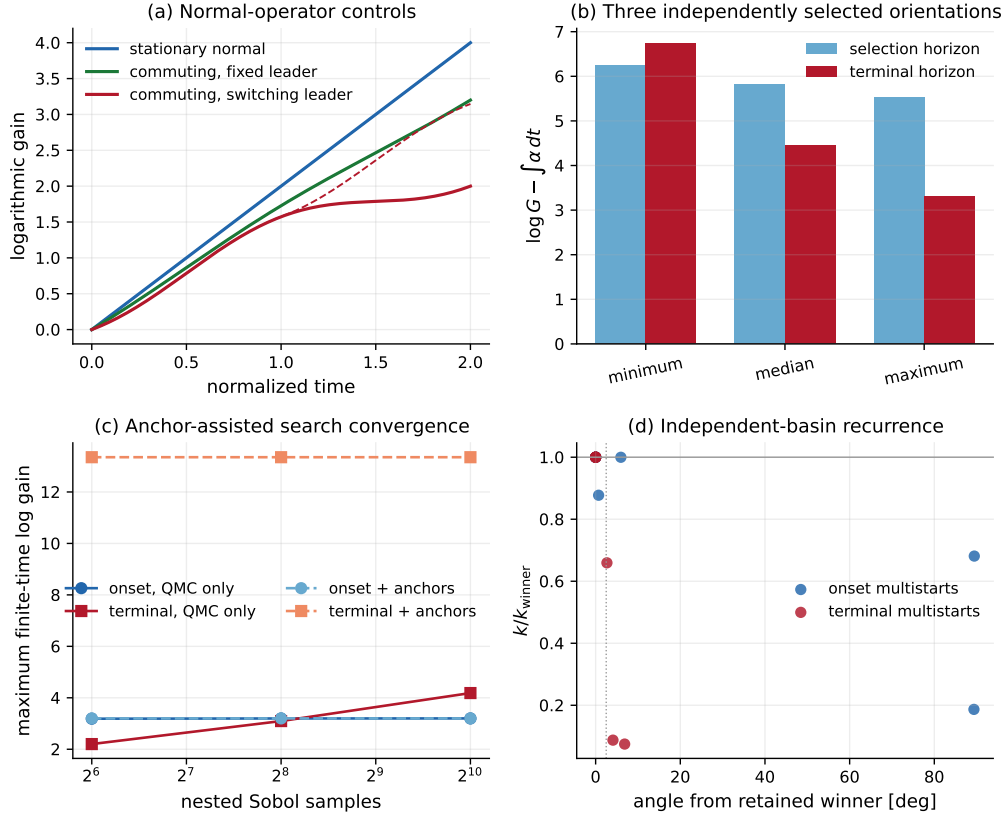


Figure 5: Strengthening audits. (a) Stationary-normal and commuting-normal fixed-leader controls give coincident propagated and frozen-integral curves; a commuting-normal switching-leader control correctly separates them. (b) Like-for-like full-state discrepancies for three independently selected crystal orientations. (c) Nested projective-sphere- $\log k$ reconnaissance, with blind Sobol and anchor-combined maxima shown separately. (d) Recurrence of eight independent local refinements around the retained onset and terminal basins. Panels (c,d) constitute an anchor-assisted basin-retention audit, not a global-optimum certificate.

5.5. Direction and wavenumber reorganize with the observation horizon

The tested gain envelope crosses $\mathcal{G}_c = e$ at

$$t_c \approx 1.46 \mu\text{s}, \quad (78)$$

using log-linear interpolation between the registered 1.4375 and 1.5 μs checkpoints. At 1.5 μs , the refined x - and y -like branches have gains 5.3141 and 5.3207. Their difference is only 0.124%, so both belong to the 2% near-optimal set. Their wavenumbers, 2.416×10^4 and $5.125 \times 10^4 \text{ m}^{-1}$, correspond to wavelengths of 260 and 123 μm . A unique onset direction or wavelength would therefore be a tie-breaking artefact.

The horizon map in Fig. 6 shows two simultaneous changes. The best tested direction rotates rapidly from an x -like to a y -like branch, while the selected k first rises and then decreases as the observation window lengthens. At 140 μs , the continuous reference optimization gives a y -like branch with $k = 7.724 \times 10^3 \text{ m}^{-1}$, wavelength 0.8135 mm and gain 6.388×10^5 . Its gain gap to the best tested competitor is 99.994% in relative terms. The selected wavelength is thus a windowed property of the evolving path, not a fixed material length.

Time refinement supports the threshold calculation. At fixed 129-state history, increasing midpoint substeps from two to four changes t_c by 2.43×10^{-6} relatively and the final log gain by 3.04×10^{-5} . The remaining 1.97% difference between the 65- and 129-state histories is controlled by base-history resolution. Independent factor-two changes of the thermal, plastic, density and signed-slip coordinate scales change t_c by at most 2.34% and preserve the qualitative branch/mechanism conclusion, but do not make t_c coordinate invariant.

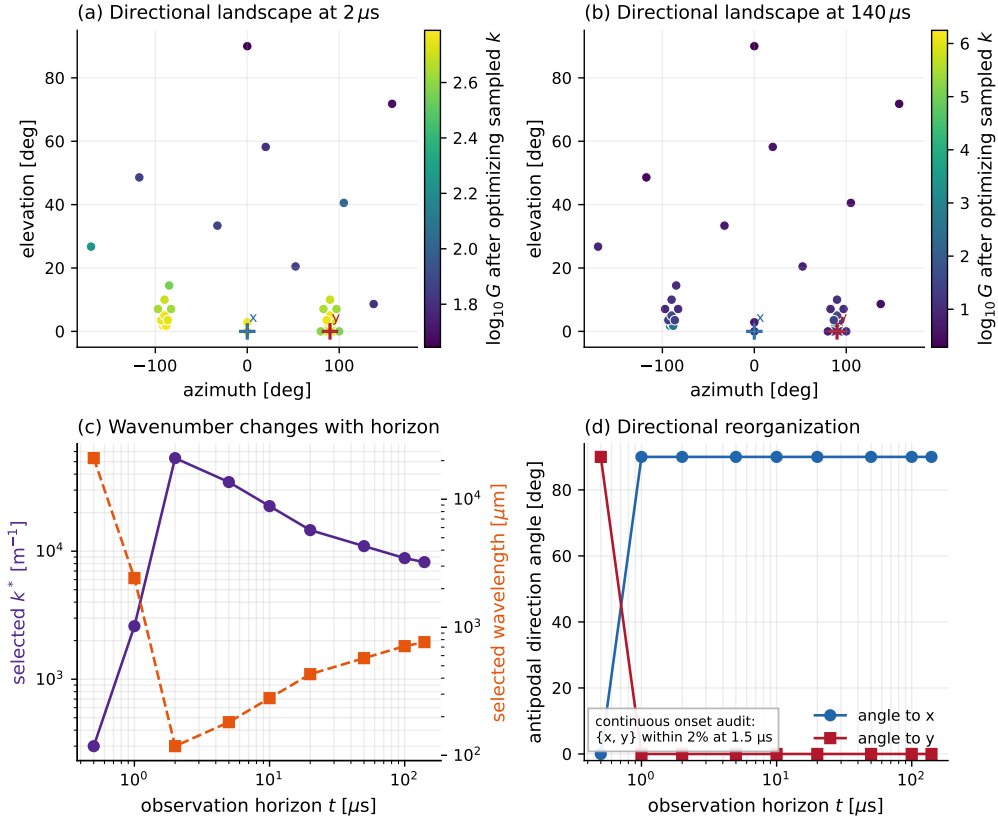


Figure 6: Direction–wavenumber selection as a function of observation horizon. Panels (a,b) show the registered directional reconnaissance at 2 and $140 \mu\text{s}$ after optimizing sampled k ; panels (c,d) track the selected wavenumber/wavelength and direction. The independent continuous onset audit retains both x - and y -like branches within 2% at $1.5 \mu\text{s}$. The landscape scan visualizes basin reorganization; the reported reference values come from the continuous branch refinement.

5.6. Selector-specific re-optimization separates localization from boundary waves

Re-optimizing direction and wavenumber for all seven selectors under the baseline and four adverse norms changes the interpretation of the unrestricted state maps (Fig. 7). Of the 70 horizon–norm–selector problems, 15 attain the upper compact-domain bound. At onset, full/full and mechanical/mechanical do so under every norm; at the terminal horizon only mechanical/mechanical does. For the baseline norm these boundary gains continue to increase at fixed direction when k is extended from 3×10^5 to $3 \times 10^7 \text{ m}^{-1}$. They are therefore high-frequency elastic/inertial wave branches, not finite localization wavelengths. Every selector that observes a constitutive or temperature response retains an interior optimum under every adverse norm. The complete fixed-candidate 11×7 terminal matrix, including the six additional factor-two norms, is reported as Fig. S1 rather than compressed into the representative main-text panel.

For the baseline constitutive/constitutive map, the re-optimized onset point is $k^* = 7.185 \times 10^4 \text{ m}^{-1}$ with $\log \mathcal{G} = 0.6217$; the 129-state screening map places the terminal point at $k^* = 6.837 \times 10^3 \text{ m}^{-1}$ with $\log \mathcal{G} = 13.3622$ ($\mathcal{G} = 6.36 \times 10^5$). Across the five norms, its onset k^* spans 6.286 – $7.185 \times 10^4 \text{ m}^{-1}$, whereas the terminal value remains within 6.8365 – $6.8369 \times 10^3 \text{ m}^{-1}$. Doubling the dislocation coordinate scale raises the terminal log gain to 14.0552 without moving that wavenumber basin. Thus the terminal constitutive localization basin is much more stable than the unrestricted full-state optimum. The subsequent fixed-direction/fixed-wavenumber density ladder supplies the central 1025-state terminal gain 6.29×10^5 ; it refines the amplification magnitude without redefining the selected basin.

The formal singular-value gate does not reject any of the 60 non-rank-one problems. For the baseline constitutive/constitutive selector, σ_1/σ_2 is 1.862 at onset and 3.73×10^5 at the terminal horizon, both above the 1.0526 degeneracy limit. The ten constitutive/temperature cases are rank-one observables, for which a second singular value and a degeneracy ratio are not defined. Near equality of two *directional basins* must therefore not be confused with singular-value degeneracy at one optimized map.

The same-selector piecewise-frozen propagator converges to the time-ordered map when every registered nonuniform interval is refined. At 16 subdivisions per interval, the maximum absolute log-gain error over all 14 baseline horizon–selector pairs is 0.0982 ; for the constitutive/constitutive map it is 0.0209 at onset and 0.00715 at the terminal horizon. Hence freezing

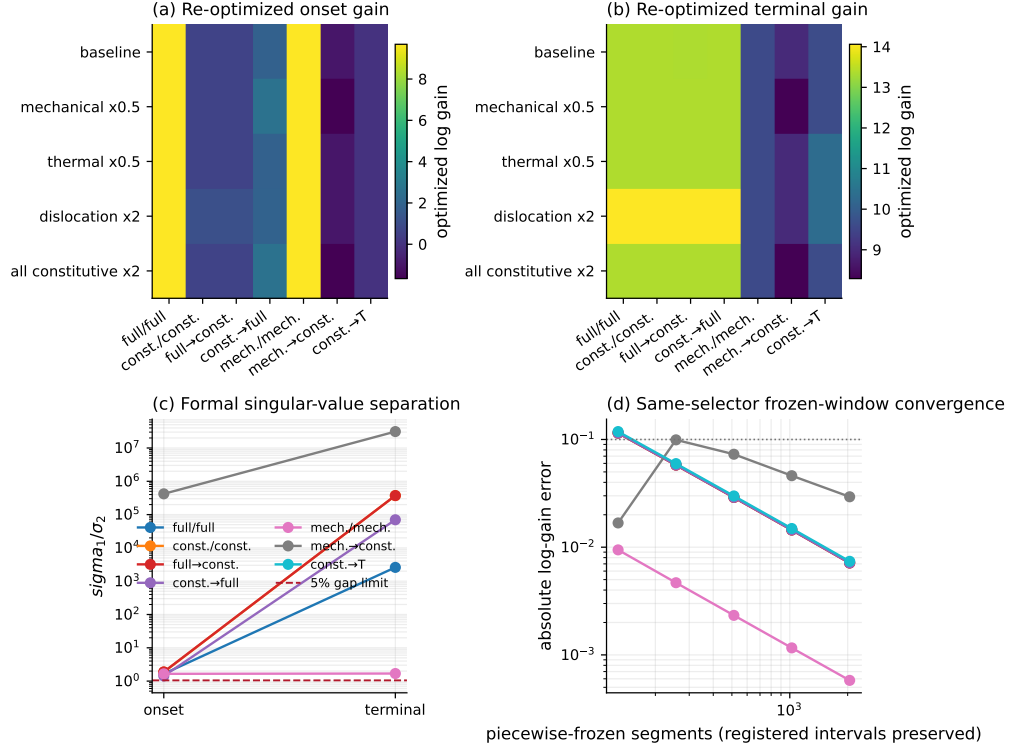


Figure 7: Selector-specific direction–wavenumber re-optimization. (a,b) Optimized log gains for seven selectors under the baseline and four adverse norms. (c) Onset and terminal singular-value ratios with the formal 5% relative-gap threshold; rank-one temperature outputs are not assigned a ratio. (d) Convergence of the same- $\mathbf{P}_{\text{in}}/\mathbf{P}_{\text{out}}$ piecewise-frozen propagator while preserving every registered nonuniform base-state interval. Boundary k maxima are classified as wave branches and are not converted into localization wavelengths.

a generator over successively shorter intervals is a consistent approximation to propagation. The failure of the scalar $\int \alpha dt$ surrogate arises from discarding vector transport and chronology, not from an unavoidable mismatch of endpoint selectors.

5.7. Rate transfer exposes scale sensitivity and experimental limits

Without refitting, the three added paths— 5.0×10^3 and $1.3 \times 10^4 \text{ s}^{-1}$ at 293 K, and $1.0 \times 10^4 \text{ s}^{-1}$ at 373 K—all preserve an interior terminal constitutive/constitutive basin. Their terminal wavenumbers are respectively 3.727×10^3 , 8.421×10^3 and $6.291 \times 10^3 \text{ m}^{-1}$ (Fig. 8), with log gains 13.6343, 13.4334 and 12.5373. The orientation changes by less than 0.01° in the local

parameter study, so the main transfer is a selected scale and amplification effect rather than a branch rotation.

The local derivatives separate parameters that control amplification from those that control scale. The mobile-dislocation seed has gain sensitivities $S_{\rho_{m0}}^{\mathcal{G}} = -0.783$ at onset and -0.918 at the terminal horizon. K_{Λ} is nearly inactive at onset (-0.009) but has the largest terminal gain sensitivity ($+1.516$). The Nye length is inactive to the displayed precision on this branch. By contrast, ℓ_{χ} has negligible onset gain sensitivity but $S_{\ell_{\chi}}^{k*} = -0.329$. A nearly unchanged gain therefore does not imply a robust selected length.

The external Grade-II CP-Ti comparison is deliberately negative as a quantitative validation. At $1.3 \times 10^4 \text{ s}^{-1}$, the model peaks at 580 MPa and $1.35 \mu\text{s}$ with only 0.224 K temperature rise. Guo et al. report an approximately 455 MPa peak at $58.5 \mu\text{s}$ and a 50–90 K rise at that peak [20]. Both histories place peak stress before the largest temperature rise, but the model misses the specimen-scale timing, temperature magnitude and post-localization stress collapse. This comparison supports only the qualitative ordering; it does not validate the verification seed as a calibrated titanium card.

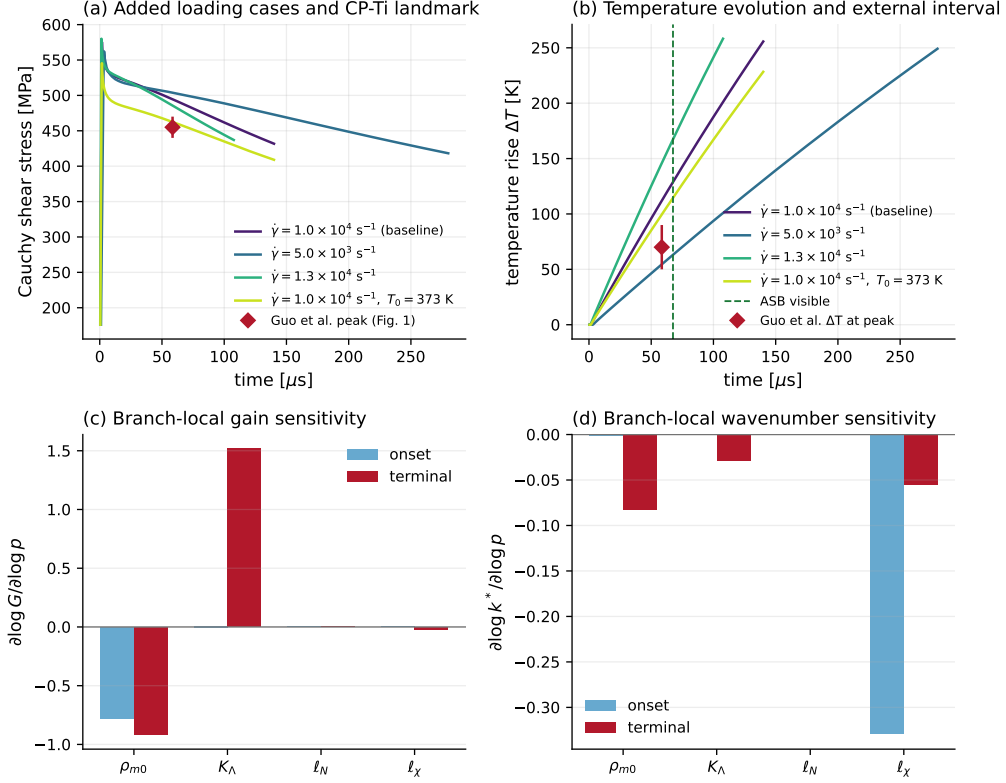


Figure 8: Transfer and validation boundaries. (a,b) Stress and temperature histories for the three added rate/temperature paths, with synchronized Grade-II CP-Ti landmarks from Guo et al. (c,d) Branch-local logarithmic sensitivities of gain and optimized wavenumber to $\pm 20\%$ changes in two dislocation and two gradient parameters. Experimental markers are external timing/scale checks, not fitted data.

5.8. The coupling controlling amplification changes with time

Participation of the final singular vector is dominated by plastic-kinematic coordinates, but participation alone does not establish causal dependence. The fixed-base intervention in Fig. 9 supplies the stronger model-level test. At 2 μ s, deleting thermal cross-links changes the log gain by only -0.013 , whereas deleting inertia cross-links changes it by -2.634 and deleting dislocation cross-links by -5.668 . At 140 μ s, the corresponding changes are -5.583 , -13.726 and -5.219 . The complete Shapley vector has cosine similarity only 0.400 between the two horizons; its leading absolute edge changes from dislocation-plastic coupling at onset to inertia-plastic coupling at the

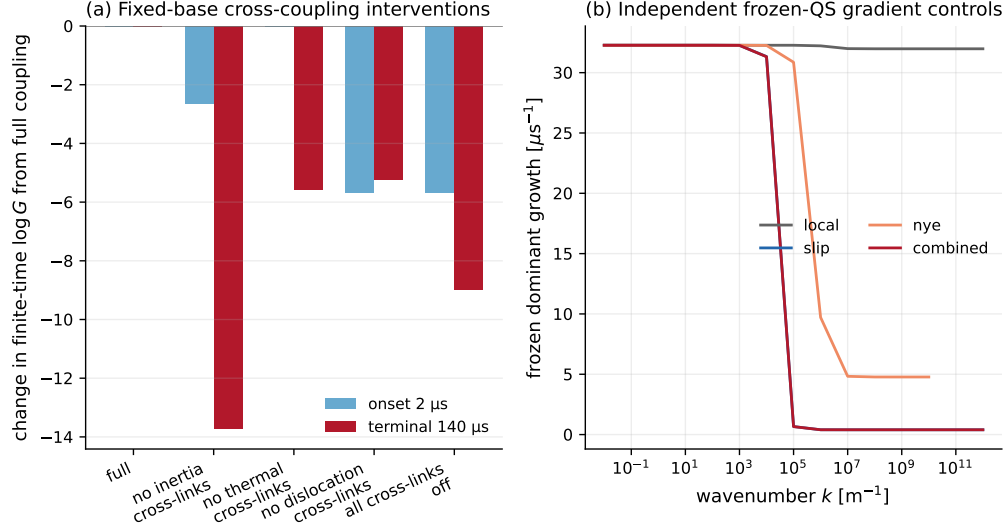


Figure 9: Counterfactual controls. (a) Change in finite-time log gain after grouped removal of cross-coupling blocks while holding the base history and all within-group blocks fixed. The dependence on inertia, thermal and dislocation cross-links changes sharply between 2 and 140 μ s. (b) Independent frozen quasi-static gradient audit: full-rank slip-gradient recovery controls the sampled high- k branch, whereas the rank-deficient Nye term alone gives only partial suppression. The panels have separate evidence contracts and are not compared quantitatively.

terminal horizon. One edge also changes attribution sign. The mechanism is therefore temporally reorganized rather than represented by one invariant label such as “thermal” or “dislocation controlled.”

The independent quasi-static gradient controls in Fig. 9(b) answer a narrower question. The local model alone retains nearly constant high- k frozen growth. Adding the Nye term reduces growth only partially because its symbol is rank deficient, whereas the full-rank slip-gradient term, alone or combined with Nye recovery, suppresses the sampled high- k branch. This establishes which regularization acts on the audited quasi-static symbol; it is not a nonlinear band-width calibration and is not quantitatively combined with panel (a).

5.9. Low-mode nonlinear transport passes, but the full-Fourier gate remains open

The density ladder is closed before nonlinear comparison. The near-onset 257-to-513 relative log-gain change is 2.615%, which triggers the mandatory 1025-state calculation; the subsequent 513-to-1025 change is 0.666%. The corresponding terminal change is 0.245%. The 1025-state four-substep map therefore supplies the central linear values $\mathcal{G} = 1.7302$ near onset and 6.2906×10^5 at the terminal horizon, with the V3 directions, wavenumbers and metric held fixed.

On the matching 513-state map, a four-nonnegative-mode Galerkin replay of the prescribed right singular vectors passes its small-amplitude checks (Fig. 10). Near onset, the linear gain 1.7365 is reproduced as 1.7238, a 0.730% difference; at 140 μs , the linear gain 6.0898×10^5 is reproduced as 6.0763×10^5 , a 0.223% difference. The complete complex output-vector errors on that Galerkin subspace are at most 1.006% and 0.223%, and factor-two amplitude changes alter the measured gain by less than 4×10^{-7} relatively.

This result cannot be promoted to complete grid-resolved nonlinear validation. When all nonnegative Fourier modes resolvable on the 16-cell grid are retained, the near-onset continuation becomes non-finite with both one and four nonlinear substeps per refined interval. The four-mode replay therefore demonstrates low-mode consistency only. It does not establish nonlinear harmonic closure, finite-amplitude saturation or a mature band width.

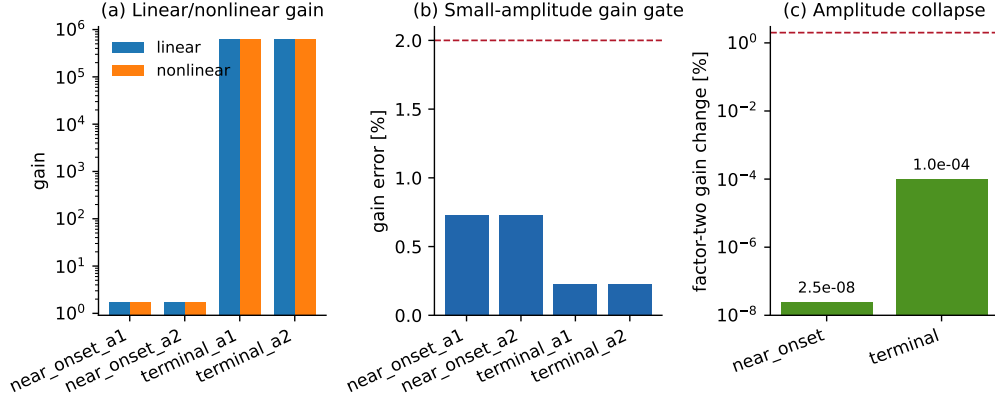


Figure 10: Four-mode Galerkin transport check for the actual near-onset and terminal optimal inputs. (a) Linear and nonlinear gains for two factor-two amplitudes at each horizon. (b) Relative gain errors remain below the preregistered 2% gate within this subspace. (c) Factor-two gain changes are orders of magnitude below the 2% amplitude-collapse gate. The separate full-Fourier audit remains open and prevents interpreting this figure as complete nonlinear validation.

5.10. The same-window bridge verifies transport but rejects a material width

The same-window bridge passes its lower-level tests (Fig. 11). Linear gain and phase translation are reproduced. A prescribed single Fourier mode remains amplitude linear over the accepted range, and its measured FWHM converges with grid refinement. However, the converged width follows the imposed wavelength:

$$0.32970 \leq W_{\text{FWHM}}/\lambda_{\text{input}} \leq 0.32979. \quad (79)$$

This near-constant ratio is evidence of faithful transport of the imposed mode, not emergence of an independent material scale.

The 27 seed-spectrum-amplitude cases fail the required seed-independent FWHM and monotonic contraction gates. The multimode peaks are resolved by only 2.55–2.93 cells, below the preregistered 12-cell requirement. Consequently, the correct decision is asymmetric: the linear/nonlinear transport bridge is verified, but the claim of an emergent material shear-band width is rejected for the present calculations. Grid convergence of a seeded single-mode width is necessary and demonstrable; it is not sufficient for physical length-scale identification.

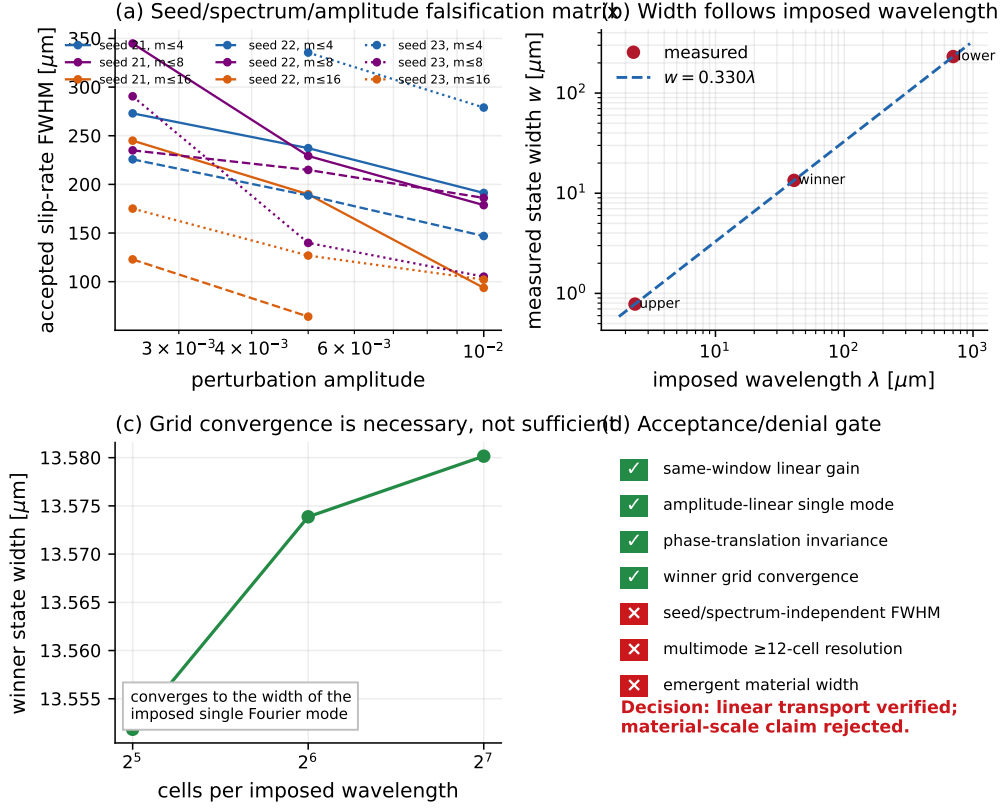


Figure 11: Same-window linear-nonlinear acceptance/denial gate. (a) The FWHM depends on perturbation amplitude, seed and spectral support. (b) Single-mode width follows $W \simeq 0.330\lambda_{\text{input}}$. (c) That imposed-mode width converges under grid refinement. (d) Gain, amplitude linearity, phase translation and winner-grid convergence pass, whereas seed-independent FWHM, 12-cell multimode resolution and an emergent material width fail.

6. Discussion

6.1. What is explained that existing theories do not explain?

The principal advance is not another instantaneous instability criterion. Existing scalar thermoplastic theories explain how hardening, rate sensitivity, thermal softening, inertia and diffusion enter a frozen dispersion relation [1, 4, 5, 3]. Frozen crystal tangents add orientation and slip-system resolution. Neither construction, by itself, answers the history-selection question posed here: among all admitted crystal-resolved initial disturbances, which

direction–wavenumber–state combination gains most over a specified interval while the base state changes?

The same-branch discrimination makes this distinction falsifiable. If accumulated frozen spectral abscissae were an adequate surrogate, $\Delta_{\text{FT}}^{\text{full}}$ in Eq. (69) would remain small when the base path, branch, 69-state space and norm are locked. Instead, the like-for-like discrepancies are +4.784, +7.217 and +6.402. The previously apparent negative onset- x discrepancy arose from subtracting a full-state frozen rate from a projected constitutive gain; separating those two input–output questions removes that artefact without removing the non-equivalence. No branch-independent shift or multiplicative calibration of the frozen rate recovers the three propagated histories.

The propagator resolves the missing chronological information. It maps every admissible initial disturbance to the terminal observed state, and its leading singular triplet identifies the initial composition, cumulative gain and terminal composition that are optimal for the declared norm and window. The measured leading-mode condition numbers reach 2.96×10^5 , adjacent eigenvectors rotate by as much as 51.1° , and adjacent generators do not commute. On the terminal branch, reversing the chronological factors lowers the log gain from 13.370 to 10.979, while the first-Magnus surrogate gives 10.115. These diagnostics identify non-orthogonal modal geometry, rotating instantaneous subspaces and operator ordering as active sources of the frozen/propagated difference. Optimizing the resulting map over projective direction and wavenumber produces a near-optimal *set*, not merely a largest root. The theory therefore explains three observations unavailable from a single frozen spectrum: why two different directions are effectively tied near onset, why one branch later separates by orders of magnitude, and why the selected wavenumber drifts from a short onset scale to a longer terminal scale.

6.2. *A mechanism is a history-dependent coupling architecture*

Labeling the dominant terminal singular-vector block as “plastic” is descriptive but incomplete. The intervention results show why. At onset, dislocation–plastic coupling carries the largest absolute Shapley attribution; at the terminal horizon, inertia–plastic coupling does. The full attribution vectors have low cosine similarity and one cross-edge changes sign. Thus the terminal plastic state can be produced through different upstream coupling architectures at different horizons. This resolves an ambiguity common in localization discussions: a coordinate with the largest amplitude need not be the coupling whose removal changes amplification most.

The fixed-base interventions are intentionally more modest than causal identification in a physical specimen. Removing a Jacobian block changes the linearized model while holding its base path fixed; it answers a counterfactual question about the tangent architecture, not how a material would re-equilibrate after a constitutive mechanism were physically removed. Nevertheless, recomputation of all 64 coalitions is stronger evidence than reading components of one eigenvector or one singular vector, and it demonstrates temporal reorganization without conflating it with base-state changes.

6.3. Relation to Bai theory and crystal plasticity

The framework should be read as a crystal-resolved generalization of the *question* posed by Bai, not yet as a completed asymptotic reduction to Bai’s coefficients. The scalar theory and the HCP descriptor have both been independently audited. What remains open is a controlled chain of assumptions—single effective slip, isotropic/specified orientation reduction, frozen internal variables, compatible thermal variables and elimination order—under which the HCP determinant reduces analytically to the classical scalar polynomial. Until that chain is written and verified, “recovers Bai exactly” would be stronger than the evidence.

This boundary does not weaken the finite-time result. The core discrimination is internal to the HCP model: the frozen and propagated calculations use the same generator, complete state space and scaling; they differ only in whether chronological operator composition is retained. Therefore the conclusion that frozen accumulation is not equivalent does not depend on a cross-model Bai calibration. Bai remains the classical anchor; the new object is the crystal-resolved nonautonomous propagator.

6.4. How far does the theory close cross-scale length and energy?

The present theory closes one important part of the cross-scale problem. Crystal orientation, slip-system kinetics, mobile/dipole dislocation evolution, heat generation, inertia, conduction and gradient recovery all enter one finite-time operator. An interior k^* for a constitutive-response selector is a finite-band selection result, and the first-law ledger partitions crystal-plastic power into heat and a passive nonthermal remainder without identifying the latter as dislocation free energy. The unrestricted full-state statement is weaker: its onset optimum is a high- k boundary wave that continues to grow when the compact search interval is extended. The selector-specific re-optimization

therefore identifies exactly which input–output questions admit a finite localization wavelength and which do not.

It does *not* yet close the full chain from grain/void scale to process-zone width, propagation resistance and structural band thickness. Four gaps remain. First, $2\pi/k^*$ is the period of an optimally amplified linear disturbance, not the width of a mature localization zone. Second, the separate same-window width audit finds that accepted single-mode widths scale with the imposed wavelength and that multimode widths are under-resolved; no seed-independent material width emerges. Third, neither an identified internal-variable Helmholtz potential nor a propagation-resistance or fracture-like energy per unit area has been derived from the volumetric power ledger. Fourth, although a four-mode Galerkin replay reproduces the optimized linear inputs, retaining every Fourier mode resolvable on the same 16-cell grid makes the nonlinear continuation non-finite even after substep refinement. Thus the current crystal-plasticity–perturbation theory closes finite-time finite-band *linear selection*; it does not close nonlinear harmonic transfer, finite width, finite dissipation or propagation resistance across scales.

The negative width result is consequential. Earlier cross-horizon FWHM/wavelength ratios can organize numerical cases, but they do not identify a constitutive transfer coefficient because their windows differ. Even under a matched window, grid convergence alone is insufficient when the converged width tracks the imposed seed. A future material-length claim must pass the complete denial gate: broadband/multimode initiation, phase and spectrum insensitivity, at least 12 cells across the peak, refinement stability, and an energy balance that yields a finite areal dissipation or propagation resistance.

6.5. Mathematical and evidential limits

The critical time in Eq. (78) is a threshold crossing in a declared dimensionless norm. Equivalent finite-dimensional norms preserve finite-time boundedness, but they do not preserve a numerical $\mathcal{G} = e$ crossing. The complete 11-by-7 fixed-candidate screen is therefore only a sensitivity map over three registered branches. The unrestricted 70-problem re-optimization supplies the stronger result: every constitutive/thermal-response selector retains an interior optimum under the five audited norms, while full/full and mechanical/mechanical can select a high- k boundary wave. This is robustness conditional on a declared physical question, not coordinate or selector

invariance. None of the 60 non-rank-one optimized maps triggers the formal singular-value degeneracy gate; the ten scalar-temperature maps have no σ_2 . This pointwise separation does not remove the set-valued near-onset competition between distinct direction–wavenumber basins.

The strengthened search combines projective-sphere–log k reconnaissance, exact symmetry and prior-branch anchors, basin recurrence, boundary classification and 129-state reevaluation. Because blind Sobol sampling can miss a narrow basin, it remains an anchor-assisted basin-retention and convergence audit, not an independent global-search certificate or a proof over every basin in $\mathbb{RP}^2 \times [k_{\min}, k_{\max}]$. The strictly same-selector piecewise-frozen control converges under interval-preserving refinement; this verifies the propagation implementation but does not turn the scalar frozen-abscissa integral into an equivalent vector propagator.

Positive sampled acoustic, conduction and gradient symbols remove obvious pathologies but do not prove a uniform high- k bound for a non-normal propagator. Proposition 5 states the missing symmetrizer condition explicitly. The current high- k calculations are consistency evidence on registered paths, not construction of that symmetrizer. Constitutive switching surfaces are excluded because ordinary branch-locked Jacobians are used; a semismooth or measure-differential extension would be required to cross them rigorously.

The failed full-Fourier continuation is consistent with this limitation and prevents using the low-mode replay as a substitute for a high- k theorem. It does not by itself distinguish physical short-wave instability from loss of continuum validity, aliasing or insufficient nonlinear regularization. Closing that distinction requires a convergent de-aliased spatial ladder and a constitutively justified high-gradient or nonlocal energy, rather than an ad hoc spectral cutoff.

Finally, the titanium-like card exercises the complete operator but is not identified against a matched batch, loading history and full-field localization measurement. The direct comparison with synchronized Grade-II CP-Ti data makes the limitation quantitative: at a matched nominal rate the model peaks far too early, produces almost no temperature rise at that peak and cannot reproduce the post-localization stress collapse. Appendix Appendix C makes the provenance distinction parameter by parameter: heat-capacity and selected thermal data have direct reference provenance, the crystal/slip block is literature constrained but largely inherited from a watermarked verification seed, and the gradient lengths are registered model

controls rather than measured band widths. Titanium experiments demonstrate that shear response, GND accumulation, alloy state and band self-organization provide material-specific constraints that a verification card cannot replace [20, 21, 22, 23]. The predicted t_c , gain and k^* should therefore be interpreted as theory-discriminating computational observables. Specimen-level prediction requires independent calibration and validation after, not before, the operator and scale gates are closed.

7. Conclusions

This study resolves a question that frozen-time localization analyses cannot answer: which admissible crystal perturbation wins after the complete thermomechanical history has acted. Five conclusions delimit the resulting advance.

1. Exact elimination of the algebraic micromorphic fields produces a 69-state finite generator without adding fictitious inertia. Generalized Schur verification and stationary/commuting-normal positive controls show that the observed departure from accumulated frozen growth is a physical consequence of non-normal, rotating and noncommuting HCP tangents, rather than a state-space, projection or implementation mismatch.
2. The answer depends on the input–output question. Across 70 selector–norm–horizon re-optimizations, unrestricted full-state and mechanical maps can select a compact-domain high-frequency wave branch that has no finite localization wavelength. Constitutive-response selectors instead retain interior maxima; the baseline terminal constitutive/constitutive map gives $k^* = 6.837 \times 10^3 \text{ m}^{-1}$ and the 1025-state map gives $\mathcal{G} = 6.29 \times 10^5$. Same-selector piecewise-frozen products converge to the time-ordered propagator, and no non-rank-one case triggers the formal 5% singular-value degeneracy gate.
3. The terminal constitutive basin transfers across three added rate/temperature paths, but both selected wavenumber and amplification change. Local sensitivity further separates amplification controls from scale controls: ρ_{m0} controls gain at both horizons, K_Λ becomes dominant at the terminal horizon, and ℓ_χ can shift onset k^* with negligible change in gain. The Grade-II titanium comparison retains only stress-before-temperature ordering and rejects quantitative agreement in peak time, temperature rise and post-localization

collapse. The present card is therefore a verification seed, not a calibrated titanium predictor.

4. Mechanism attribution is time dependent: the leading cross-coupling changes from dislocation–plastic near onset to inertia–plastic at the terminal horizon, while full-rank slip-gradient recovery controls the sampled quasi-static high- k branch more completely than the rank-deficient Nye term. A four-mode Galerkin replay transports both re-optimized singular inputs within its small-amplitude acceptance gates. Retaining every resolvable Fourier mode instead makes the continuation non-finite even after substep refinement. The evidence therefore validates finite-time linear selection and low-mode consistency, but not complete nonlinear transport or finite-amplitude saturation.
5. The closed result is therefore finite-time, selector-dependent linear selection on an evolving HCP path. The same-window audit still rejects a seed-independent mature width, and the present theory does not supply specimen calibration, propagation resistance, fracture-like dissipation or a strict HCP-to-Bai analytical degeneration. The open full-Fourier gate further shows that finite-band linear selection alone does not regularize nonlinear harmonic transfer. Complete generators, singular subspaces, optimization traces, parameter provenance and run-time contracts make those open gates explicit and reproducible.

The central distinction is consequently sharper than a new instability number: a finite-time propagator can identify which crystal perturbation is amplified by an evolving path, while the present evidence also states when that perturbation cannot be interpreted as a material shear-band width.

Data and code availability

The public reproducibility repository is available at <https://github.com/yxlnbu/finite-time-hcp-perturbation>. It contains the machine-readable base histories, all 129 complex generators for every reported baseline branch, complete singular vectors and spectra, full-state/projected operator receipts, the fixed 11-by-7 matrix, selector-specific optimization traces, piecewise-frozen controls, loading and parameter-transfer tables, Galerkin replay fields, the 1025-state density receipt, the failed full-Fourier audit, literature-digitization landmarks, parameter provenance and the locked run-time environment. V3 contracts are recorded in `05_results/ijp_revisio`

n_evidence_v3.json, 05_results/ijp_v3_validation_suite_v1.json and 05_results/ijp_v3_full_fourier_nonlinear_audit_v1.json; release checksums and array contracts are recorded in 05_results/ijp_reproducibility_release_v1.json. Selection of a reuse license and deposition under a persistent DOI remain to be completed before submission.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work: **[to be confirmed by all authors].**

Funding

[Funding sources and grant numbers to be supplied.]

CRediT authorship contribution statement

[Author-specific CRediT roles to be supplied.]

Appendix A. Block reduction and finite-time bounds

Appendix A.1. Exact algebraic elimination

With

$$\mathbf{Q}_{yy}(s) = \mathbf{K}_{yy} + s\mathbf{E}_{yy} + s^2\mathbf{M}_{yy}, \quad (\text{A.1})$$

the augmented pencil is

$$\mathcal{Q}(s) = \begin{bmatrix} \mathbf{Q}_{yy}(s) & \mathbf{K}_{y\zeta} \\ \mathbf{K}_{\zeta y} & \mathbf{K}_{\zeta\zeta} \end{bmatrix}. \quad (\text{A.2})$$

If $\mathbf{K}_{\zeta\zeta}$ is invertible, introduce

$$\mathbf{L} = \begin{bmatrix} \mathbf{I} & -\mathbf{K}_{y\zeta}\mathbf{K}_{\zeta\zeta}^{-1} \\ \mathbf{0} & \mathbf{I} \end{bmatrix}, \quad \mathbf{R}_e = \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ -\mathbf{K}_{\zeta\zeta}^{-1}\mathbf{K}_{\zeta y} & \mathbf{I} \end{bmatrix}. \quad (\text{A.3})$$

Both matrices have determinant one, and direct multiplication gives

$$\mathbf{L}\mathcal{Q}(s)\mathbf{R}_e = \begin{bmatrix} \mathbf{Q}_{yy}(s) - \mathbf{K}_{y\zeta}\mathbf{K}_{\zeta\zeta}^{-1}\mathbf{K}_{\zeta y} & \mathbf{0} \\ \mathbf{0} & \mathbf{K}_{\zeta\zeta} \end{bmatrix}. \quad (\text{A.4})$$

This proves the determinant factorization in Eq. (35). If $\hat{\mathbf{y}}$ belongs to the reduced kernel, then $[\hat{\mathbf{y}}, -\mathbf{K}_{\zeta\zeta}^{-1}\mathbf{K}_{\zeta y}\hat{\mathbf{y}}]^\top$ belongs to the augmented kernel. Conversely, the lower row of Eq. (A.2) forces this reconstruction for every augmented kernel vector. Thus the finite algebraic and geometric multiplicities are preserved.

Appendix A.2. Generator equivalence and consistent initialization

Write the reduced quadratic equations as

$$\mathbf{R}\ddot{\mathbf{u}} + \mathbf{K}_{uu}\mathbf{u} + \mathbf{K}_{ud}\mathbf{d} = \mathbf{0}, \quad (\text{A.5})$$

$$\mathbf{C}\dot{\mathbf{d}} + \mathbf{K}_{du}\mathbf{u} + \mathbf{K}_{dd}\mathbf{d} = \mathbf{0}. \quad (\text{A.6})$$

With $\mathbf{v} = \dot{\mathbf{u}}$, the first-order generator is

$$\mathbf{A}_{\text{dyn}} = \begin{bmatrix} \mathbf{0} & \mathbf{I} & \mathbf{0} \\ -\mathbf{R}^{-1}\mathbf{K}_{uu} & \mathbf{0} & -\mathbf{R}^{-1}\mathbf{K}_{ud} \\ -\mathbf{C}^{-1}\mathbf{K}_{du} & \mathbf{0} & -\mathbf{C}^{-1}\mathbf{K}_{dd} \end{bmatrix}. \quad (\text{A.7})$$

For a modal solution $\mathbf{q} = \hat{\mathbf{q}}e^{st}$, the first block row gives $\hat{\mathbf{v}} = s\hat{\mathbf{u}}$. Substitution into the remaining rows recovers Eqs. (A.5)–(A.6). Conversely, every reduced quadratic eigenvector with finite s defines a generator eigenvector by adjoining $s\hat{\mathbf{u}}$. This proves Proposition 2.

A physical initial perturbation must satisfy the algebraic reconstruction

$$\hat{\boldsymbol{\zeta}}_0 = -\mathbf{K}_{\zeta\zeta}^{-1}(t_0; k, \mathbf{n})\mathbf{K}_{\zeta y}(t_0; k, \mathbf{n})\hat{\mathbf{y}}_0. \quad (\text{A.8})$$

Arbitrary independent initialization of $\boldsymbol{\zeta}$ would violate the microforce constraint and is not admitted.

Appendix A.3. Propagator existence, continuity, and norm change

If $\mathbf{A}_{\text{dyn}}(t; p)$ is piecewise continuous in time and uniformly bounded for parameter $p = (k, \mathbf{n})$ on a compact admitted set, the integral equation

$$\boldsymbol{\Phi}(t, t_0; p) = \mathbf{I} + \int_{t_0}^t \mathbf{A}_{\text{dyn}}(\tau; p)\boldsymbol{\Phi}(\tau, t_0; p) d\tau \quad (\text{A.9})$$

has a unique absolutely continuous solution. Gronwall's inequality gives

$$\|\boldsymbol{\Phi}(t, t_0; p)\|_2 \leq \exp\left(\int_{t_0}^t \|\mathbf{A}_{\text{dyn}}(\tau; p)\|_2 d\tau\right). \quad (\text{A.10})$$

For two parameters p and p' , subtraction of their Volterra equations and a second Gronwall estimate yields

$$\|\Phi(t, t_0; p) - \Phi(t, t_0; p')\|_2 \leq C_I \int_{t_0}^t \|\mathbf{A}_{\text{dyn}}(\tau; p) - \mathbf{A}_{\text{dyn}}(\tau; p')\|_2 d\tau, \quad (\text{A.11})$$

where C_I depends only on the uniform generator bound and the time interval. Continuity of the generator in p therefore implies continuity of the propagator and of its largest singular value. Compactness then gives the maximizer in Proposition 4.

For two fixed diagonal scalings $\mathbf{D}_1$ and $\mathbf{D}_2$, define $\mathbf{S} = \mathbf{D}_2 \mathbf{D}_1^{-1}$. Their scaled propagators satisfy

$$\hat{\Phi}_2 = \mathbf{S}^{-1} \hat{\Phi}_1 \mathbf{S}. \quad (\text{A.12})$$

Consequently,

$$\kappa_2(\mathbf{S})^{-1} \|\hat{\Phi}_1\|_2 \leq \|\hat{\Phi}_2\|_2 \leq \kappa_2(\mathbf{S}) \|\hat{\Phi}_1\|_2. \quad (\text{A.13})$$

Boundedness is preserved by a fixed change of norm, but a prescribed threshold time need not be.

Appendix A.4. Conditional high-wavenumber estimate

Let $\mathbf{W}(t; k, \mathbf{n})$ be Hermitian and suppose that, for $k \geq k_0$,

$$c_- \mathbf{I} \preceq \mathbf{W} \preceq c_+ \mathbf{I}, \quad \dot{\mathbf{W}} + \mathbf{W} \hat{\mathbf{A}} + \hat{\mathbf{A}}^H \mathbf{W} \preceq 2\beta \mathbf{W}, \quad (\text{A.14})$$

with constants independent of t , k , and $\mathbf{n}$. For $\mathcal{E} = \hat{\mathbf{q}}^H \mathbf{W} \hat{\mathbf{q}}$,

$$\dot{\mathcal{E}} \leq 2\beta \mathcal{E}, \quad \mathcal{E}(t) \leq e^{2\beta(t-t_0)} \mathcal{E}(t_0). \quad (\text{A.15})$$

Assume additionally that the selector maps have uniformly finite energy constants

$$\|\mathbf{P}_{\text{out}} \mathbf{x}\|_2^2 \leq C_{\text{out}} \mathbf{x}^H \mathbf{W} \mathbf{x}, \quad (\text{A.16})$$

$$\inf_{\mathbf{P}_{\text{in}} \mathbf{x} = \mathbf{y}} \mathbf{x}^H \mathbf{W} \mathbf{x} \leq C_{\text{in}} \|\mathbf{y}\|_2^2. \quad (\text{A.17})$$

Then every selected input has a lift whose propagated output satisfies

$$\|\mathbf{P}_{\text{out}} \Phi(t, t_0) \mathbf{P}_{\text{in}}^T\|_2 \leq \sqrt{C_{\text{in}} C_{\text{out}}} e^{\beta(t-t_0)}. \quad (\text{A.18})$$

This is an observable bound; it does not assert uniform Euclidean control of every unobserved displacement and velocity component. The numerical principal-symbol checks establish necessary admissibility diagnostics but do not construct $\mathbf{W}$ for the full HCP generator.

Appendix B. Computational reproducibility and claim gates

The claims and figures are tied to the following machine-readable artefacts in the project package:

1. `bai_1982_scalar_regression_reference_v5_independent_redteam_v1.json`: scoped scalar Bai coefficient/root regression and explicit non-authority for the HCP limit;
2. `cp_ti_dynamic_descriptor_qz_v1.json`: 87-coordinate generalized Schur audit, finite/infinite root classification, complete finite-root matching and regularization-denial gates;
3. `cp_ti_finite_time_dynamic_perturbation_v2.json`: base trajectory, operator contract, coordinate scales and original direction-wavenumber scan;
4. `cp_ti_continuous_spectrum_robustness_v2.json`: 129-state propagation reference, continuously refined branches, temporal convergence, norm sensitivity, principal-symbol checks and sampled tail;
5. `cp_ti_finite_time_discrimination_v1.json`: historical projected-gain diagnostic retained for traceability and not used for the primary like-for-like subtraction;
6. `ijp_operator_consistency_v2.json` and `ijp_operator_consistency_v2_arrays.npz`: unified full-state propagator/frozen comparisons, projected positive controls, logarithmic-norm bounds, commutators, modal condition numbers, eigenvector rotations and the complete norm-selector sensitivity matrix;
7. `ijp_strengthening_evidence_v1.json`: stationary/commuting-normal controls, three-orientation transfer and an anchor-assisted projective-sphere-wavenumber basin-retention audit with complete optimizer trajectories;
8. `ijp_strengthening_evidence_v1_arrays.npz`: all complex 69-by-69 generator checkpoints, propagators, input/output singular vectors and singular spectra for every reported baseline branch, plus three-orientation generators and complete global samples;
9. `cp_ti_mechanism_causality_v2.json`: all 64 fixed-base cross-coupling coalitions, exact Shapley allocations and temporal-reorganization diagnostics;
10. `hcp_qs_spectrum_verification_seed_v1.json`: independent frozen quasi-static gradient controls;

11. `cp_ti_fwhm_seed_spectrum_v1.json` and `cp_ti_lsbrp_q3_nonautonomous_bridge_v1.json`: same-window seed/spectrum/amplitude matrix, linear transport checks, grid convergence and nonlinear material-scale denial;
12. `ijp_singular_vector_nonlinear_validation_v2.json`, `ijp_singular_vector_nonlinear_validation_v2.csv` and `ijp_singular_vector_nonlinear_validation_v2_fields.npz`: historical four-mode periodic replay of the pre-V3 near-onset and terminal right singular vectors, retained for traceability rather than complete nonlinear authority;
13. `ijp_revision_evidence_v3.json`, `ijp_revision_evidence_v3_arrays.npz` and `ijp_revision_evidence_v3/`: selector-specific direction-wavenumber re-optimization under five norms, formal singular-value-gap diagnostics, same-selector piecewise-frozen controls, three loading transfers, four local parameter sensitivities, and the bounded commercial-purity titanium comparison;
14. `ijp_singular_vector_nonlinear_validation_v3.json`, `ijp_singular_vector_nonlinear_validation_v3.csv` and `ijp_singular_vector_nonlinear_validation_v3_fields.npz`: four-mode Galerkin replay of the re-optimized baseline constitutive/constitutive onset and terminal singular vectors, including the failed discarded-RHS Fourier-energy gate;
15. `ijp_v3_dense_convergence_v1.json` and `ijp_v3_validation_suite_v1.json`: mandatory 513-to-1025 density refinement after the 257-to-513 trigger, together with the combined open/pass gate ledger;
16. `ijp_v3_full_fourier_nonlinear_audit_v1.json`: one- and four-substep attempts retaining every Fourier mode resolvable on the 16-cell grid, both terminating non-finitely near onset and therefore leaving the complete nonlinear gate open;
17. `ijp_core_figure_set_v1.json`: one receipt connecting every core figure to source artefacts and explicit claim boundaries.
18. `ijp_material_parameter_provenance_v1.csv`: 74-row parameter-unit-source-evidence-class table covering every field of the active material object and explicitly listing inactive twinning and numerical controls;
19. `ijp_reproducibility_release_v1.json`: archive checksum, manifest checksum, array-shape contract, CRC result and environment-lock

verification for 04_reproducibility/ijp_finite_time_release_v1.zip.

The corresponding scripts include `build_ijp_operator_consistency_v2.py`, `build_ijp_strengthening_evidence_v1.py`, `build_ijp_revision_evidence_v3.py`, `run_ijp_singular_vector_nonlinear_validation_v3.py`, `run_ijp_v3_dense_convergence_v1.py`, `audit_ijp_v3_full_fourier_nonlinear_v1.py`, `build_ijp_material_parameter_provenance_v1.py`, `build_ijp_reproducibility_release_v1.py` and `build_ijp_core_figure_set_v1.py`. The numerical gates require: (i) the conditioning limits in Eq. (63); (ii) exact finite/infinite descriptor root counts; (iii) finite-root and reconstruction errors below registered tolerances; (iv) convergence under independent base-history and propagator refinement; (v) less than 10% change in t_c under registered factor-two scale tests; (vi) like-for-like full-state comparison and invariant-projection positive-control equality below 2×10^{-11} ; (vii) stability across the registered norm-selector matrix and three orientations; (viii) selector-specific direction-wavenumber re-optimization with boundary maxima classified rather than converted into wavelengths; (ix) the formal singular-value separation gate in Eq. (61); (x) convergence of the same- $\mathbf{P}_{\text{in}}/\mathbf{P}_{\text{out}}$ piecewise-frozen control; (xi) a converged 1025-state linear gain and matching-grid Galerkin replay, with complete nonlinear interpretation separately conditioned on the full-Fourier gate; and (xii) explicit failure labels for unmatched experimental landmarks, full-Fourier continuation and unresolved or seed-dependent nonlinear widths. A failed gate disables the corresponding interpretation rather than replacing it with a regularized value.

The local publication-ready archive now contains the complete complex optimal right/left singular vectors, all 129 generator checkpoints for every reported baseline branch, full-state and projected operator receipts, the norm-selector matrix, mechanism histories, branch-start metadata, every optimizer evaluation, nonlinear Fourier fields and the executable environment manifest. The strict HCP-to-Bai analytical reduction, an independent global-optimum proof, coordinate-invariant t_c , experimental onset validation, an identified dislocation free energy, propagation resistance and material band-width identification remain false claim gates. The search evidence applies only as an anchor-assisted basin-retention audit on the registered compact domain, anchors, tolerances and discretization ladder stated in Section 4.

Appendix C. Complete material, model and numerical parameter provenance

Computational completeness does not imply material calibration. The evidence-class column distinguishes direct reference functions, grouped literature constraints, repository verification values, registered model controls and numerical controls. Parameters marked inactive are retained by the material object but do not enter the reported no-twinning branch.

Group	Symbol	Parameter	Value	Unit	Source	Evidence/calibration status
crystal	c/a	HCP lattice ratio	1.587	–	S1	L2 grouped literature constraint; literature-constrained, not batch identified
crystal	T_ref	elastic reference temperature	293.15	K	S5	L2 repository baseline; verification seed
elasticity	C_11	HCP stiffness C11	1.624e+11	Pa	S1/S5	L2 grouped literature constraint; research baseline, not batch identified
elasticity	C_12	HCP stiffness C12	9.2e+10	Pa	S1/S5	L2 grouped literature constraint; research baseline, not batch identified
elasticity	C_13	HCP stiffness C13	6.9e+10	Pa	S1/S5	L2 grouped literature constraint; research baseline, not batch identified
elasticity	C_33	HCP stiffness C33	1.807e+11	Pa	S1/S5	L2 grouped literature constraint; research baseline, not batch identified
elasticity	C_44	HCP stiffness C44	4.67e+10	Pa	S1/S5	L2 grouped literature constraint; research baseline, not batch identified
elasticity	C_66	derived HCP stiffness	3.52e+10	Pa	S7	derived; exactly derived from C11 and C12
thermal	rho_0	mass density	4510	kg m ⁻³	S1/S5	L2 grouped literature constraint; research baseline
thermal	c_p (298.15 K)	specific heat used at reference state	527.2485158	J kg ⁻¹ K ⁻¹	S2	L1 direct reference function; literature function, not fitted
thermal	A	Shomate coefficient A	5.406172	cal mol ⁻¹ K ⁻¹	S2	L1 direct reference function; admitted only on 298–700 K
thermal	B	Shomate coefficient B	4.538229	cal mol ⁻¹ K ⁻¹	S2	L1 direct reference function; admitted only on 298–700 K
thermal	C	Shomate coefficient C	-4.346881	cal mol ⁻¹ K ⁻¹	S2	L1 direct reference function; admitted only on 298–700 K
thermal	D	Shomate coefficient D	1.69235	cal mol ⁻¹ K ⁻¹	S2	L1 direct reference function; admitted only on 298–700 K
thermal	E	Shomate coefficient E	-0.034287	cal mol ⁻¹ K ⁻¹	S2	L1 direct reference function; admitted only on 298–700 K
thermal	M_Ti	titanium molar mass	0.047867	kg mol ⁻¹	S2	L1 reference constant; reference constant
thermal	kappa	thermal conductivity at 300 K	21.9	W m ⁻¹ K ⁻¹	S3	L1 selected reference value; literature value, not batch identified
thermal	beta	Taylor–Quinney heat fraction	0.9	–	S5	L2 repository baseline; not independently identified
slip kinetics	b*bas	burgers m by family: basal<a>	2.95e-10	m	S4/S5	L2 repository verification baseline; not family-by-family calibrated for the reported specimen
slip kinetics	b*pri	burgers m by family: prismatic<a>	2.95e-10	m	S4/S5	L2 repository verification baseline; not family-by-family calibrated for the reported specimen

Group	Symbol	Parameter	Value	Unit	Source	Evidence/calibration status
slip kinetics	b^{pyr}	burgers m by family: pyramidal<c+a>	5.53e-10	m	S4/S5	L2 repository verification baseline; not family-by-family calibrated for the reported specimen
slip kinetics	tau₀^{bas}	tau0 Pa: basal<a>	80000000	Pa	S4/S5	L2 repository verification baseline; not family-by-family calibrated for the reported specimen
slip kinetics	tau₀^{pri}	tau0 Pa: prismatic<a>	1.1e+08	Pa	S4/S5	L2 repository verification baseline; not family-by-family calibrated for the reported specimen
slip kinetics	tau₀^{pyr}	tau0 Pa: pyramidal<c+a>	2.2e+08	Pa	S4/S5	L2 repository verification baseline; not family-by-family calibrated for the reported specimen
slip kinetics	tau_{cut}^{bas}	tau cut Pa: basal<a>	5.5e+08	Pa	S4/S5	L2 repository verification baseline; not family-by-family calibrated for the reported specimen
slip kinetics	tau_{cut}^{pri}	tau cut Pa: prismatic<a>	5.5e+08	Pa	S4/S5	L2 repository verification baseline; not family-by-family calibrated for the reported specimen
slip kinetics	tau_{cut}^{pyr}	tau cut Pa: pyramidal<c+a>	7e+08	Pa	S4/S5	L2 repository verification baseline; not family-by-family calibrated for the reported specimen
slip kinetics	Delta F^{bas}	activation energy J: basal<a>	1.8e-19	J	S4/S5	L2 repository verification baseline; not family-by-family calibrated for the reported specimen
slip kinetics	Delta F^{pri}	activation energy J: prismatic<a>	1.8e-19	J	S4/S5	L2 repository verification baseline; not family-by-family calibrated for the reported specimen
slip kinetics	Delta F^{pyr}	activation energy J: pyramidal<c+a>	1.8e-19	J	S4/S5	L2 repository verification baseline; not family-by-family calibrated for the reported specimen
slip kinetics	v₀^{bas}	reference velocity m s: basal<a>	4000	m s ⁻¹	S4/S5	L2 repository verification baseline; not family-by-family calibrated for the reported specimen
slip kinetics	v₀^{pri}	reference velocity m s: prismatic<a>	4000	m s ⁻¹	S4/S5	L2 repository verification baseline; not family-by-family calibrated for the reported specimen
slip kinetics	v₀^{pyr}	reference velocity m s: pyramidal<c+a>	4000	m s ⁻¹	S4/S5	L2 repository verification baseline; not family-by-family calibrated for the reported specimen
slip kinetics	p^{bas}	p: basal<a>	0.7	–	S4/S5	L2 repository verification baseline; not family-by-family calibrated for the reported specimen
slip kinetics	p^{pri}	p: prismatic<a>	0.7	–	S4/S5	L2 repository verification baseline; not family-by-family calibrated for the reported specimen
slip kinetics	p^{pyr}	p: pyramidal<c+a>	0.7	–	S4/S5	L2 repository verification baseline; not family-by-family calibrated for the reported specimen
slip kinetics	q^{bas}	q: basal<a>	1.06	–	S4/S5	L2 repository verification baseline; not family-by-family calibrated for the reported specimen

Group	Symbol	Parameter	Value	Unit	Source	Evidence/calibration status
slip kinetics	q^{pri}	q: prismatic<a>	1.06	–	S4/S5	L2 repository verification baseline; not family-by-family calibrated for the reported specimen
slip kinetics	q^{pyr}	q: pyramidal<c+a>	1.06	–	S4/S5	L2 repository verification baseline; not family-by-family calibrated for the reported specimen
slip kinetics	rho_m0^{bas}	rho mobile 0 m2: basal<a>	1.6e+12	m ⁻²	S4/S5	L2 repository verification baseline; not family-by-family calibrated for the reported specimen
slip kinetics	rho_m0^{pri}	rho mobile 0 m2: prismatic<a>	1.6e+12	m ⁻²	S4/S5	L2 repository verification baseline; not family-by-family calibrated for the reported specimen
slip kinetics	rho_m0^{pyr}	rho mobile 0 m2: pyramidal<c+a>	1.6e+12	m ⁻²	S4/S5	L2 repository verification baseline; not family-by-family calibrated for the reported specimen
slip kinetics	rho_d0^{bas}	rho dipole 0 m2: basal<a>	1e+10	m ⁻²	S4/S5	L2 repository verification baseline; not family-by-family calibrated for the reported specimen
slip kinetics	rho_d0^{pri}	rho dipole 0 m2: prismatic<a>	1e+10	m ⁻²	S4/S5	L2 repository verification baseline; not family-by-family calibrated for the reported specimen
slip kinetics	rho_d0^{pyr}	rho dipole 0 m2: pyramidal<c+a>	1e+10	m ⁻²	S4/S5	L2 repository verification baseline; not family-by-family calibrated for the reported specimen
dislocation	d_g	grain size	2e-05	m	S4/S5	L2 repository verification baseline; not batch identified
dislocation	K_Lambda	mean-free-path coefficient	33	–	S4/S5	L2 repository verification baseline; not batch identified
dislocation	mu_ref	reference shear modulus	4.67e+10	Pa	S4/S5	L2 repository verification baseline; not batch identified
dislocation	h_self	forest self interaction	1	–	S4/S5	L2 repository verification baseline; not batch identified
dislocation	h_lat	forest latent interaction	1.35	–	S4/S5	L2 repository verification baseline; not batch identified
dislocation	a_T	Taylor coefficient	0.3	–	S4/S5	L2 repository verification baseline; not batch identified
dislocation	c_d	minimum dipole distance	3	b	S4/S5	L2 repository verification baseline; not batch identified
dislocation	nu_c	climb attempt frequency	20000	s ⁻¹	S4/S5	L2 repository verification baseline; not batch identified
dislocation	Q_c	climb activation energy	1.2e-19	J	S4/S5	L2 repository verification baseline; not batch identified
micromorphic / gradient	H_chi	micromorphic penalty modulus	4.67e+10	Pa	S6	L3 registered model control; not identified as a material band width
micromorphic / gradient	ell_N	Nye/GND energetic length	1e-06	m	S6	L3 registered model control; not identified as a material band width
micromorphic / gradient	ell_chi	full-rank slip-gradient length	2.5e-07	m	S6	L3 registered model control; not identified as a material band width
loading	g_1	Bunge orientation case 1	0/0/0	degree	S6	L3 registered orientation case; theoretical orientation, not an experimental ODF
loading	g_2	Bunge orientation case 2	30/90/30	degree	S6	L3 registered orientation case; theoretical orientation, not an experimental ODF

Group	Symbol	Parameter	Value	Unit	Source	Evidence/calibration status
loading	g_3	Bunge orientation case 3	90/60/30	degree	S6	L3 registered orientation case; theoretical orientation, not an experimental ODF
loading	T_0	initial temperature	293.15	K	S6	L3 registered loading; prescribed
loading	dot_gamma_0	macroscopic simple-shear rate	10000	s ⁻¹	S1/S6	L2 dynamic loading constraint; representative, not specimen-matched
loading	gamma_f	final simple shear	1.4	–	S6	L3 registered loading; prescribed
numerical	rho_floor	density protection floor	10000	m ⁻²	S5	N1 numerical control; not material data
numerical	eps_det	determinant tolerance	1e-08	–	S5	N1 numerical control; not material data
numerical	Delta_gamma_max	maximum slip increment	0.0002	–	S5	N1 numerical control; not material data
numerical	Delta_f_tw_max	maximum twin increment	0.0002	–	S5	N1 numerical control; not material data
inactive	gamma_tw	twin shear override	0	–	S5	I inactive repository parameter; inactive in all reported calculations
twinning	tau_tw	twin CRSS	80000000	Pa	S5	I inactive repository parameter; inactive in all reported calculations
inactive	dot_f_tw0	twin reference rate	2500	s ⁻¹	S5	I inactive repository parameter; inactive in all reported calculations
twinning	m_tw	twin rate exponent	3	–	S5	I inactive repository parameter; inactive in all reported calculations
inactive	tau_tw_scale	twin stress scale	1.5e+08	Pa	S5	I inactive repository parameter; inactive in all reported calculations
twinning	H_tw	twin latent hardening	3e+08	Pa	S5	I inactive repository parameter; inactive in all reported calculations
inactive	f_tw_max	maximum twin fraction	0.85	–	S5	I inactive repository parameter; inactive in all reported calculations

Source key. **S1:** Nemat-Nasser, Guo and Cheng, Acta Materialia 47 (1999) 3705–3720, doi:10.1016/S1359-6454(99)00203-7; grouped alpha-Ti crystal/dynamic-loading constraint.

S2: NIST-JANAF alpha-Ti table / NIST Chemistry WebBook Shomate heat-capacity coefficients, admitted over 298–700 K.

S3: Touloukian/TPRC selected thermal-conductivity value for annealed high-purity titanium near 300 K, as registered in the material card.

S4: Dai, L. and Song, W., A strain rate and temperature-dependent crystal plasticity model for hexagonal close-packed (HCP) materials: Application to alpha-titanium, International Journal of Plasticity 154 (2022) 103281, doi:10.1016/j.ijplas.2022.103281 (publisher record accessed 2026-08-29). The present card adopts the model class only; its numerical entries were not copied or batch-calibrated parameter by parameter from that paper.

S5: Public release commit 8aa42e3b5999adc502142f940e42b9222cdb17e6, config/verification_seed.yaml (frozen kernel version 0.1, SHA-256 b7ced9294e7d17ba64b5c2eaf843be5eb4defbcfe480bbb04bce1fb8612c124a) and config/cp_ti_grade1_dynamic_hcp_v1.json (schema V1, SHA-256 586dd4ed43509aa045f9fe87524b2df51fbd07817853c497ac8c6ad4c082df5d), accessed 2026-08-29. The seed remains watermarked VERIFICATION_SEED_NOT_CALIBRATED_FOR_TA2.

S6: Present-study registered modeling choice in the finite-time spectral runner; not a measured shear-band width.

S7: Exact definition, derived quantity, or universal physical constant.

The full machine-readable table, including the model role and active/inactive flag for every row, is supplied as `ijp_material_parameter_provenance_v1.csv`.

References

- [1] Y. L. Bai, Thermo-plastic instability in simple shear, Journal of the Mechanics and Physics of Solids 30 (4) (1982) 195–207. doi:10.1016/0022-5096(82)90029-1.
- [2] Y. L. Bai, B. Dodd, Adiabatic Shear Localization: Occurrence, Theories and Applications, Pergamon Press, Oxford, 1992.
- [3] T. W. Wright, The Physics and Mathematics of Adiabatic Shear Bands, Cambridge University Press, Cambridge, 2002.

- [4] T. W. Wright, R. C. Batra, The initiation and growth of adiabatic shear bands, *International Journal of Plasticity* 1 (3) (1985) 205–212. doi:10.1016/0749-6419(85)90003-8.
- [5] A. Molinari, R. J. Clifton, Analytical characterization of shear localization in thermoviscoplastic materials, *Journal of Applied Mechanics* 54 (4) (1987) 806–812. doi:10.1115/1.3173121.
- [6] A. Benallal, D. Bigoni, Effects of temperature and thermo-mechanical couplings on material instabilities and strain localization of inelastic materials, *Journal of the Mechanics and Physics of Solids* 52 (3) (2004) 725–753. doi:10.1016/S0022-5096(03)00118-2.
- [7] M. Arriaga, H. Waisman, Combined stability analysis of phase-field dynamic fracture and shear band localization, *International Journal of Plasticity* 96 (2017) 81–119. doi:10.1016/j.ijplas.2017.04.018.
- [8] A. B. Jacquey, H. Rattez, M. Veveakis, Strain localization regularization and patterns formation in rate-dependent plastic materials with multi-physics coupling, *Journal of the Mechanics and Physics of Solids* 152 (2021) 104422. doi:10.1016/j.jmps.2021.104422.
- [9] G. Franz, F. Abed-Meraim, M. Berveiller, Strain localization analysis for single crystals and polycrystals: Towards microstructure–ductility linkage, *International Journal of Plasticity* 48 (2013) 1–33. doi:10.1016/j.ijplas.2013.02.001.
- [10] Z. Zhang, D. E. Eakins, F. P. E. Dunne, On the formation of adiabatic shear bands in textured HCP polycrystals, *International Journal of Plasticity* 79 (2016) 196–216. doi:10.1016/j.ijplas.2015.12.004.
- [11] W. An, C.-z. Liu, Q.-l. Xiong, Z. Li, X. Huang, T. Suo, Shear localization in polycrystalline metal at high-strain rates with dynamic recrystallization: Crystal plasticity modeling and texture effect, *International Journal of Plasticity* 165 (2023) 103616. doi:10.1016/j.ijplas.2023.103616.
- [12] J. Li, I. Romero, J. Segurado, Development of a thermo-mechanically coupled crystal plasticity modeling framework: Application to polycrystalline homogenization, *International Journal of Plasticity* 119 (2019) 313–330. doi:10.1016/j.ijplas.2019.04.008.

- [13] L. Dai, W. Song, A strain rate and temperature-dependent crystal plasticity model for hexagonal close-packed (HCP) materials: Application to alpha-titanium, *International Journal of Plasticity* 154 (2022) 103281. doi:10.1016/j.ijplas.2022.103281.
- [14] M. E. Gurtin, A gradient theory of single-crystal viscoplasticity that accounts for geometrically necessary dislocations, *Journal of the Mechanics and Physics of Solids* 50 (1) (2002) 5–32. doi:10.1016/S0022-5096(01)00104-1.
- [15] M. E. Gurtin, A theory of grain boundaries that accounts automatically for grain misorientation and grain-boundary orientation, *Journal of the Mechanics and Physics of Solids* 56 (2) (2008) 640–662. doi:10.1016/j.jmps.2007.05.002.
- [16] P. R. M. van Beers, G. J. McShane, V. G. Kouznetsova, M. G. D. Geers, Grain boundary interface mechanics in strain gradient crystal plasticity, *Journal of the Mechanics and Physics of Solids* 61 (12) (2013) 2659–2679. doi:10.1016/j.jmps.2013.08.011.
- [17] A. Alipour, S. Reese, S. Wulfinghoff, A grain boundary model for gradient-extended geometrically nonlinear crystal plasticity: Theory and numerics, *International Journal of Plasticity* 118 (2019) 17–35. doi:10.1016/j.ijplas.2019.01.009.
- [18] V. Phalke, S. Forest, H.-J. Chang, A. Roos, Adiabatic shear banding in FCC metallic single and poly-crystals using a micromorphic crystal plasticity approach, *Mechanics of Materials* 169 (2022) 104288. doi:10.1016/j.mechmat.2022.104288.
- [19] A. Lame Jouybari, S. El Shawish, L. Cizelj, Enhanced strain gradient crystal plasticity theory: Evolution of the length scale during deformation, *International Journal of Plasticity* 190 (2025) 104351. doi:10.1016/j.ijplas.2025.104351.
- [20] Y. Guo, Q. Ruan, S. Zhu, Q. Wei, H. Chen, J. Lu, B. Hu, X. Wu, Y. Li, D. Fang, Temperature rise associated with adiabatic shear band: Causality clarified, *Physical Review Letters* 122 (1) (2019) 015503. doi:10.1103/PhysRevLett.122.015503.

- [21] C. Zhu, V. Livescu, T. Harrington, O. Dippo, G. T. Gray, K. S. Vecchio, Investigation of the shear response and geometrically necessary dislocation densities in shear localization in high-purity titanium, *International Journal of Plasticity* 92 (2017) 148–163. doi:10.1016/j.ijplas.2017.03.009.
- [22] C. Ran, Q. Zhou, P. Chen, Q. Chen, W. Zhang, Comparative experimental study of the dynamic properties and adiabatic shear susceptibility of titanium alloys, *European Journal of Mechanics - A/Solids* 85 (2021) 104137. doi:10.1016/j.euromechsol.2020.104137.
- [23] Q. Xue, M. A. Meyers, V. F. Nesterenko, Self-organization of shear bands in titanium and Ti-6Al-4V alloy, *Acta Materialia* 50 (3) (2002) 575–596. doi:10.1016/S1359-6454(01)00356-1.